

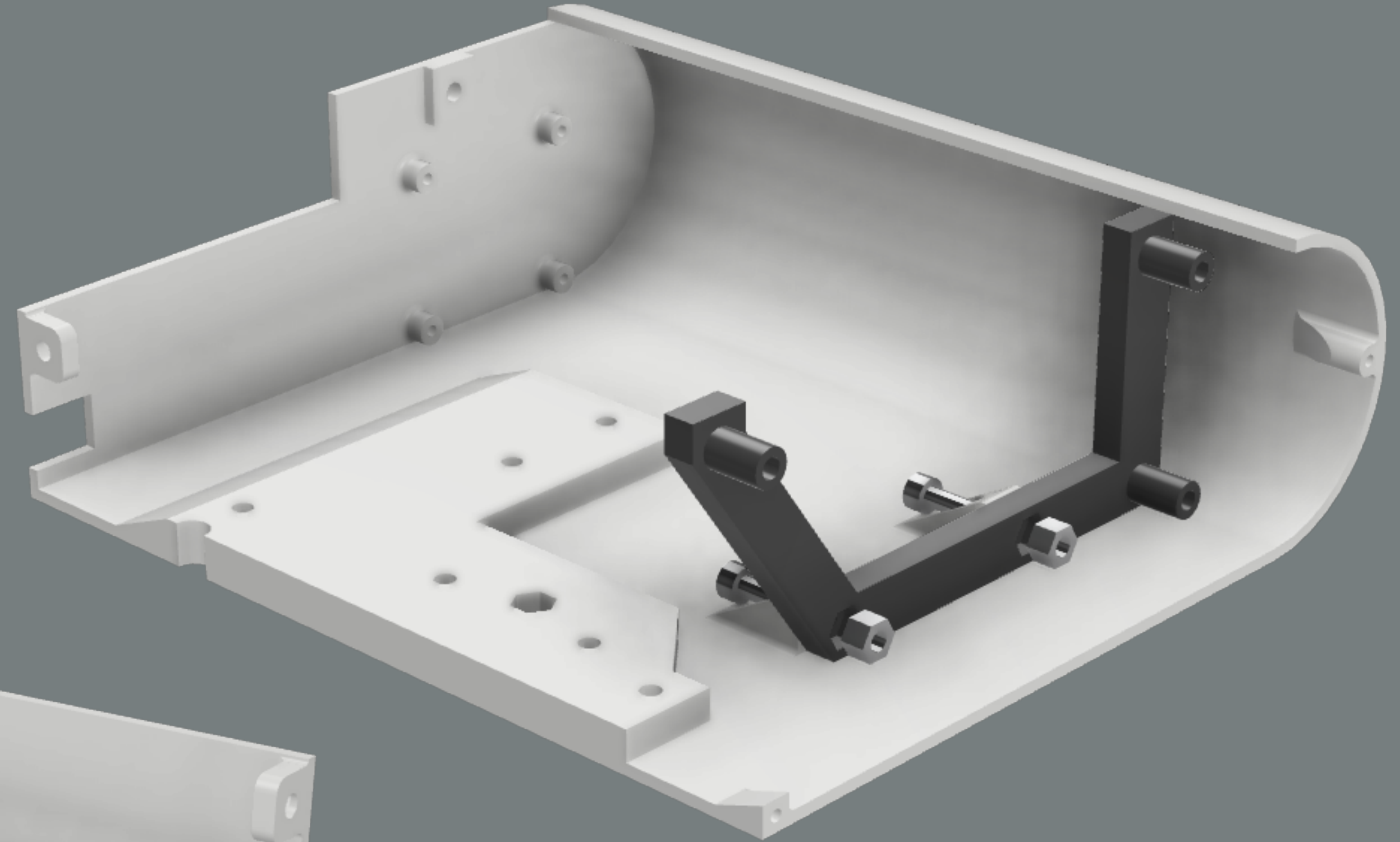
Talk Buddy Kit Build Instructions



OMINOUS INDUSTRIES 2024

Using **(4)M3x10 Screws** and **(4)M3 Nuts**, attach the two Speaker Mounts to either side case piece.

Step 1



***Make sure the direction of the two Speaker Mounts is correct**

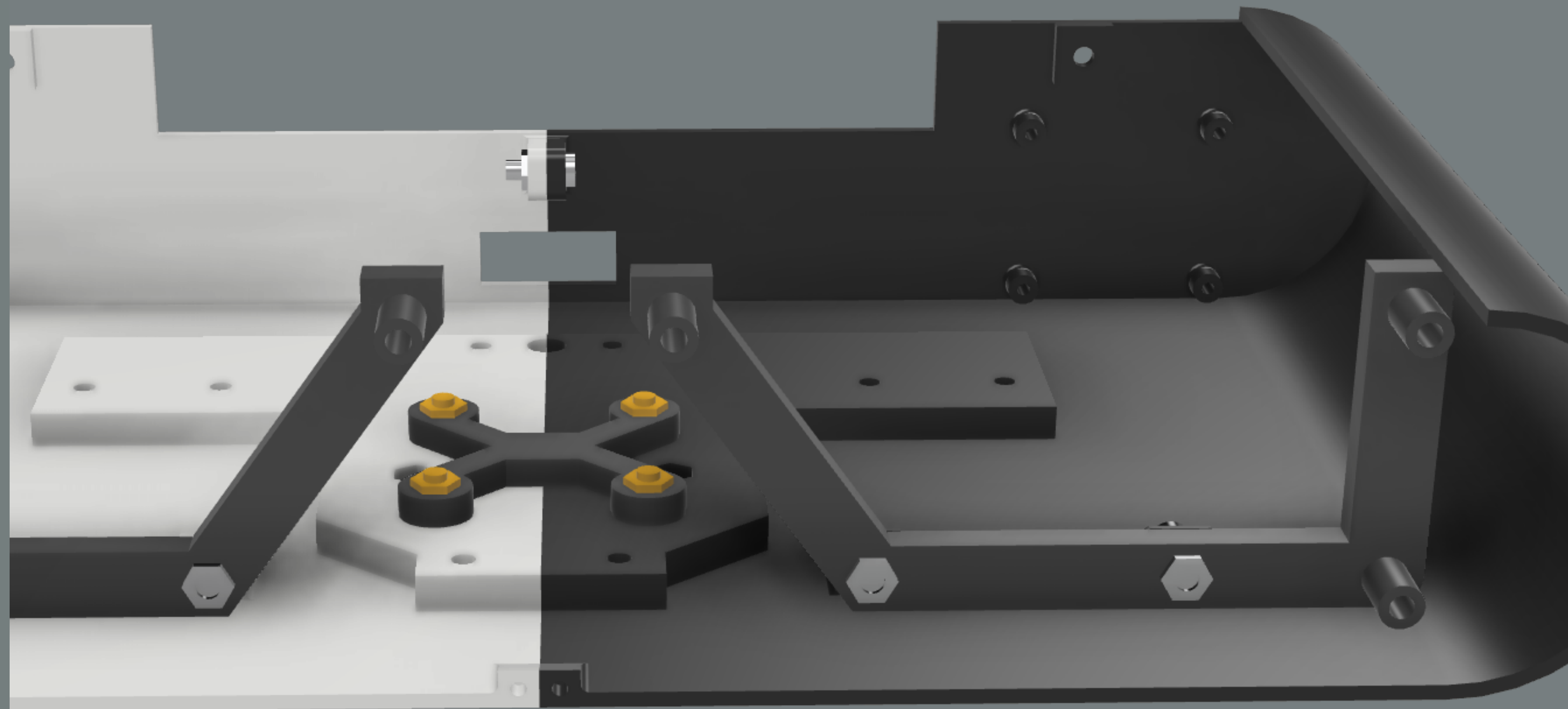
2.5mm Hex Key



Step 2

Mount the Crossmember to both case pieces using **(4)M3x10 Screws** and **(4)M3 Nuts**. Do not fully tighten yet.

If you are building the Arduino version, please stop and read the **Arduino Setup PDF** before going further.

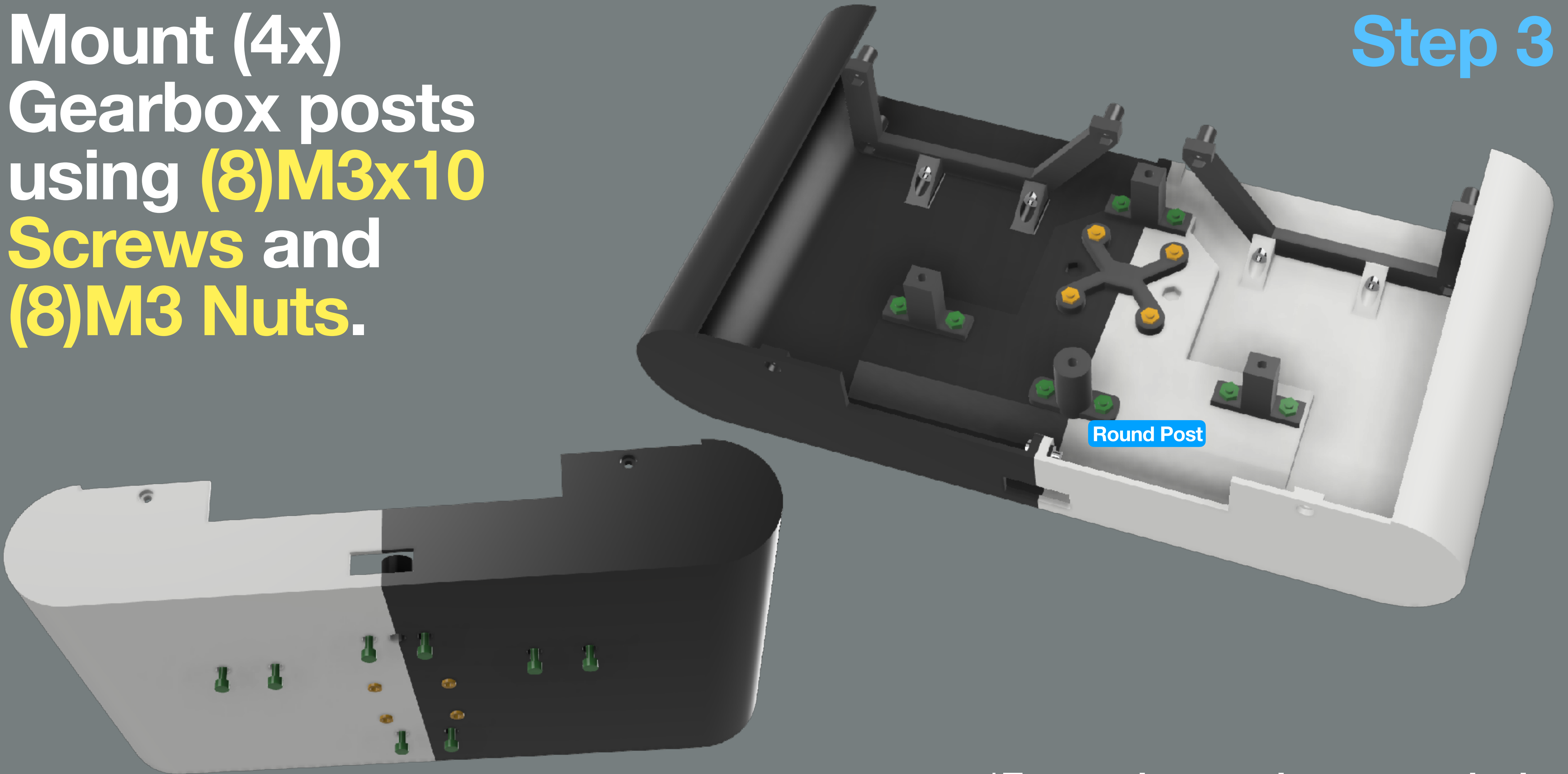


2.5mm Hex Key



Mount (4x)
Gearbox posts
using (8)M3x10
Screws and
(8)M3 Nuts.

Step 3



Round Post

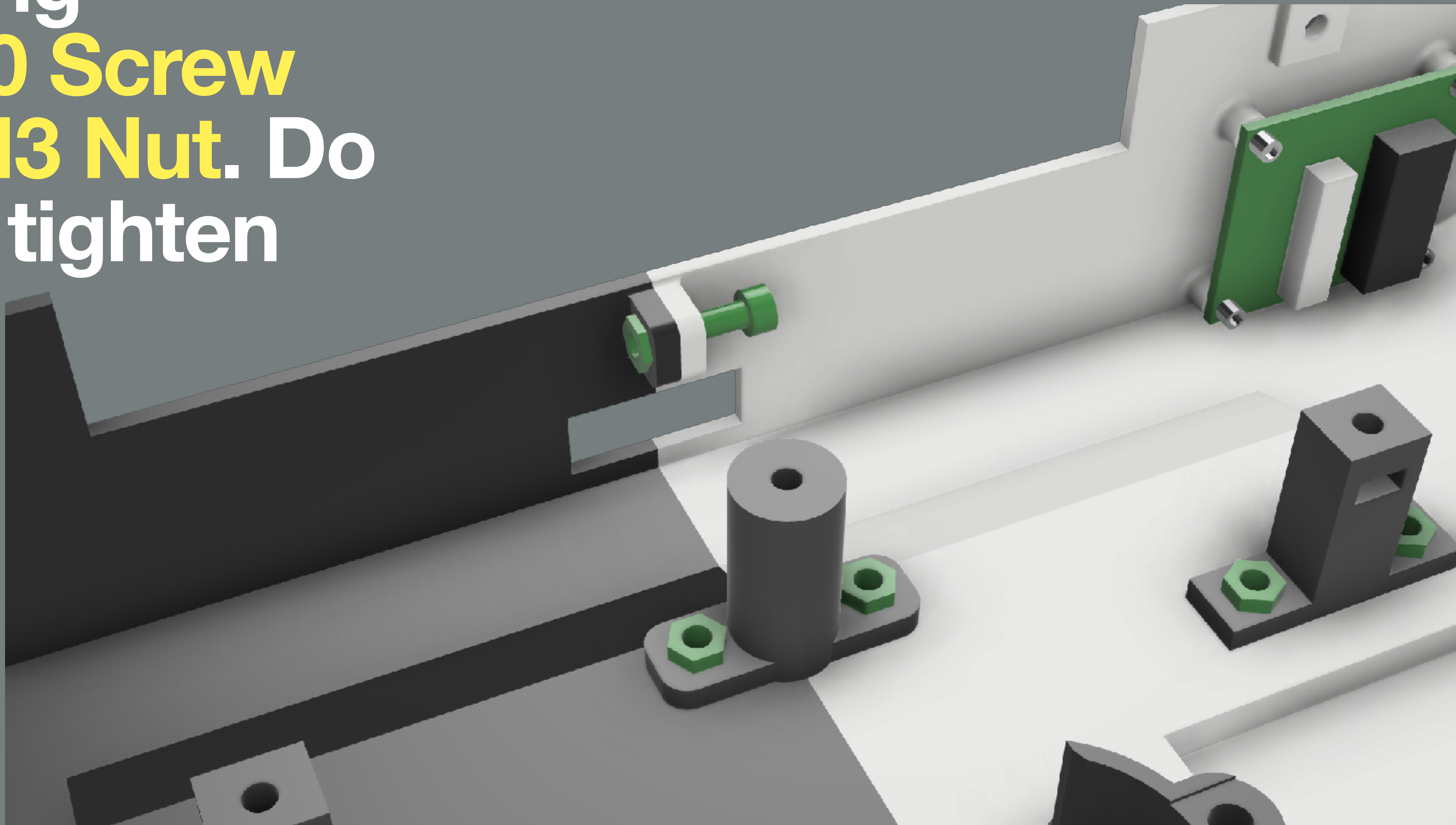
2.5mm Hex Key



*Ensure the round post goes in the rear center position of the case. Do not fully tighten yet.

Join the rear case
tabs using
(1) M3x10 Screw
and **(1) M3 Nut**. Do
not fully tighten
yet.

Step 4

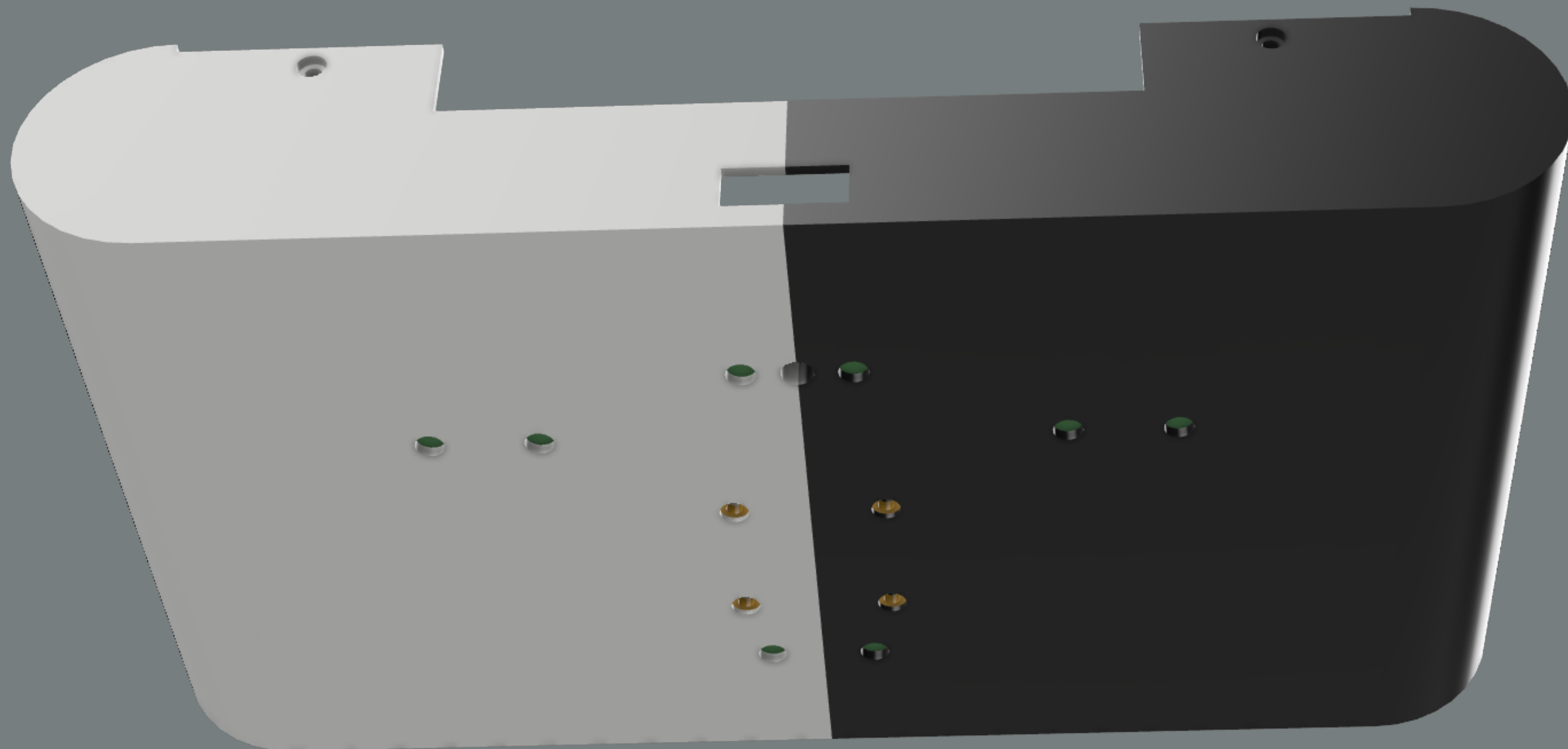


2.5mm Hex Key



Step 5

Ensure the two side case pieces are aligned properly, then proceed to tighten all the Screws to securely join the two side case pieces.

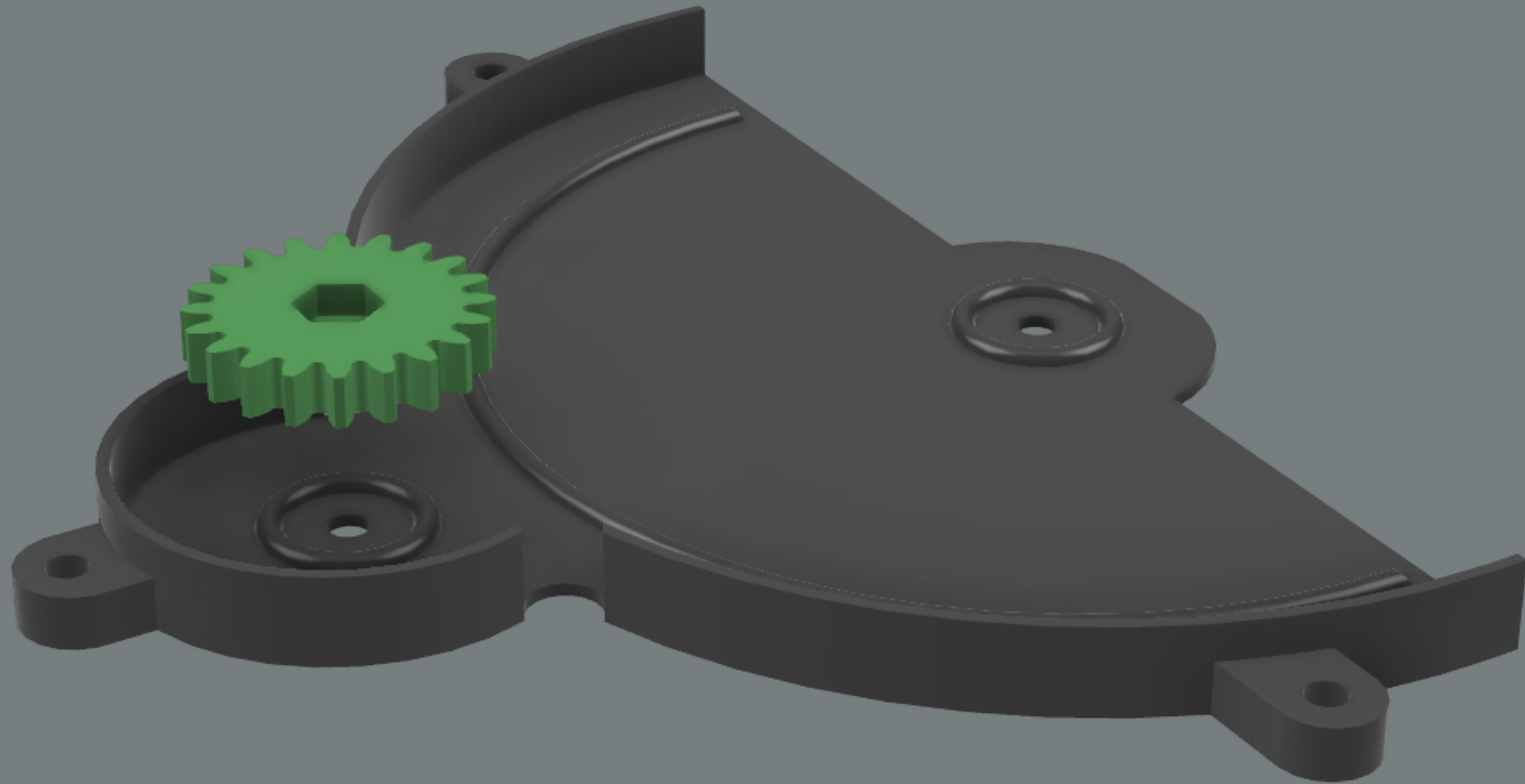


2.5mm Hex Key



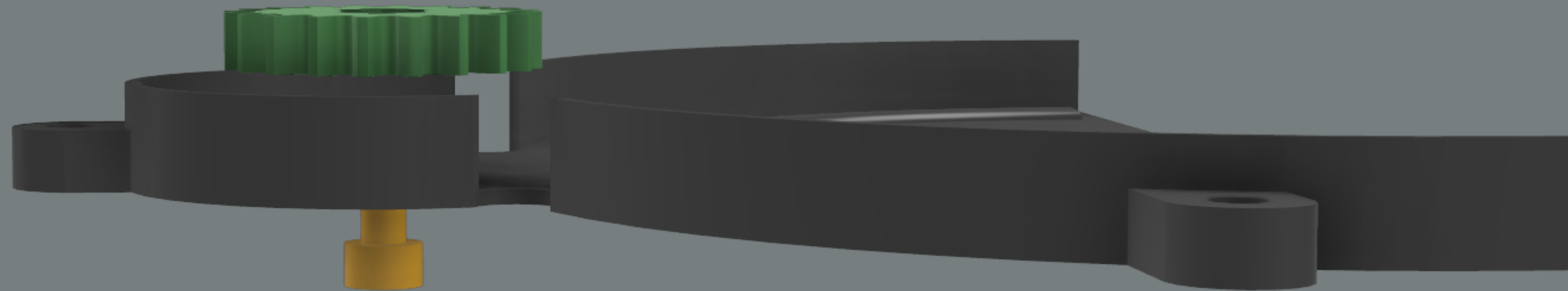
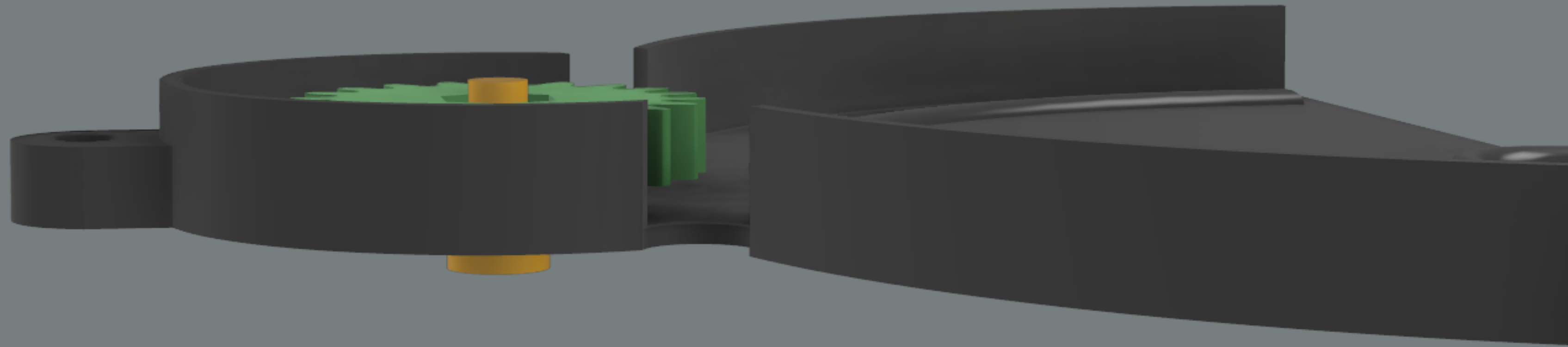
**Ready the
Gearbox
Bottom and
Small Gear.**

Step 6



Insert (1) **M3x8 Screw** through the Gearbox Bottom and the Small Gear.

Step 7



Step 8

Place (1) **M3 Lock Nut** into the recess on the Small Gear and tighten over the **M3x8 Screw** until the Small Gear is held securely on the Gearbox Bottom.

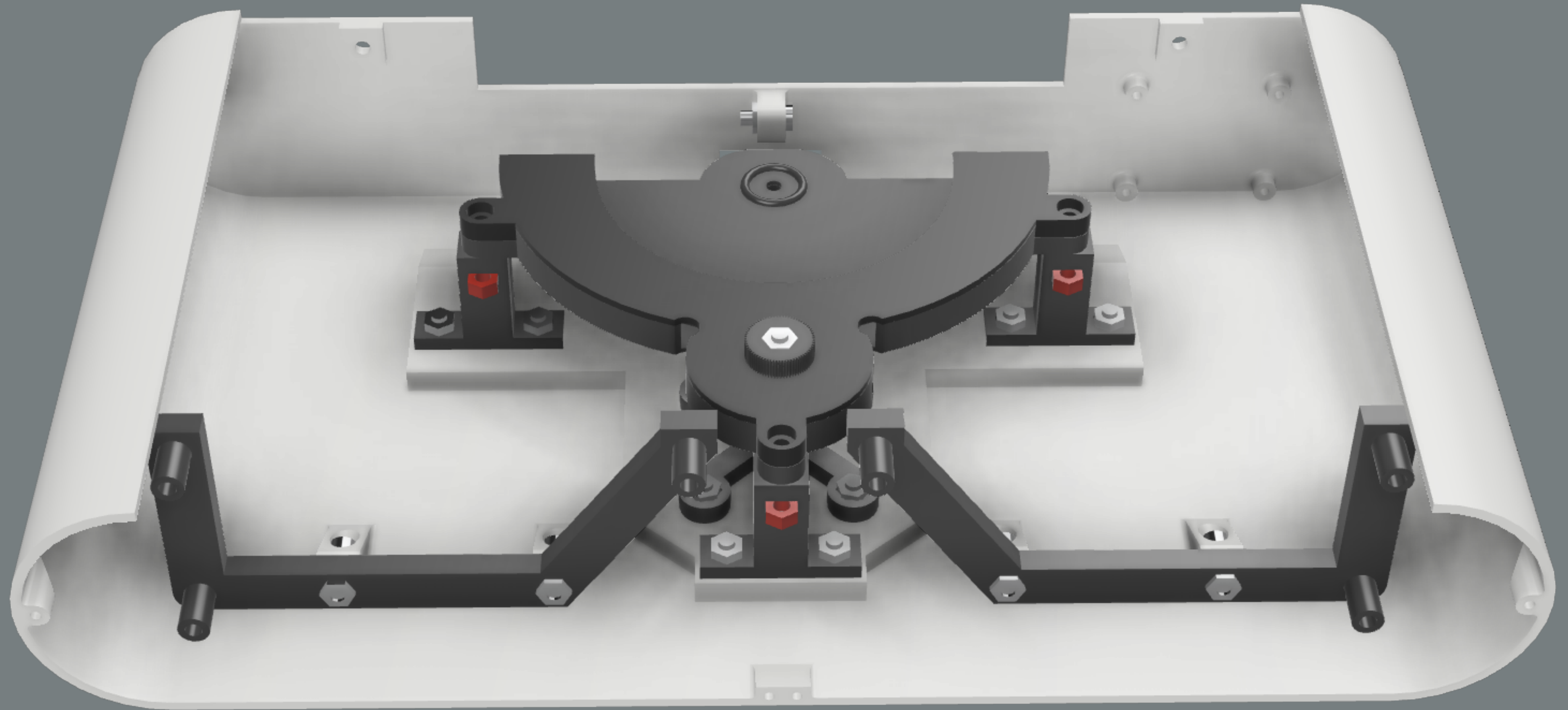


**** Do not over tighten this or the gear will be too difficult to turn. You may substitute the lock nut for a regular M3 Nut, but will need to re-tighten it occasionally as the gear movement will cause it to loosen over time.**

2.5mm Hex Key

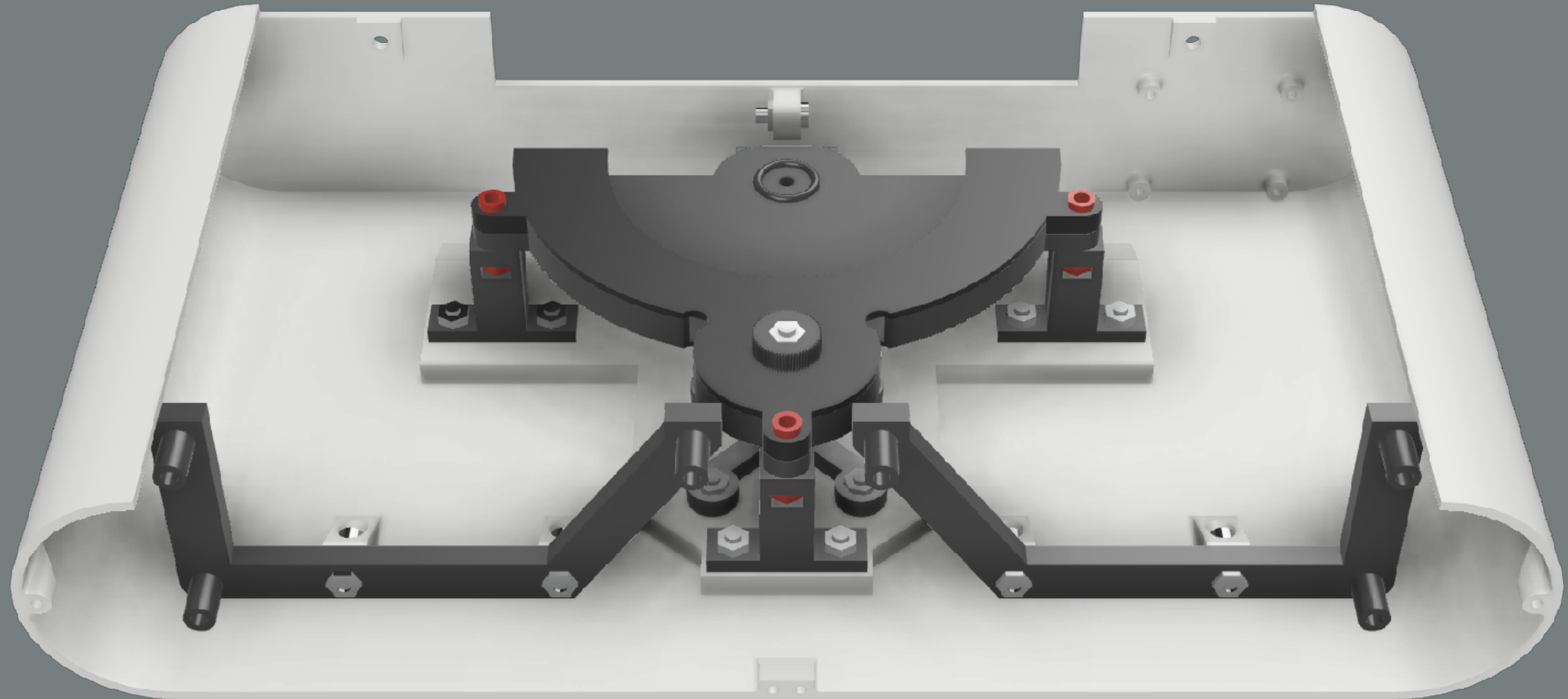


Align and place the Gearbox Assembly onto the (4x) Gearbox posts and insert **(3)M3 Nuts** into the holes in the fronts of the 3x square Gearbox Posts.



Using **(3)M3x16 Screws**, secure the Gearbox Top through the Gearbox Bottom, and into the **(3)M3 Nuts** inserted within the 3x square gearbox posts in **Step 9**.

Step 10

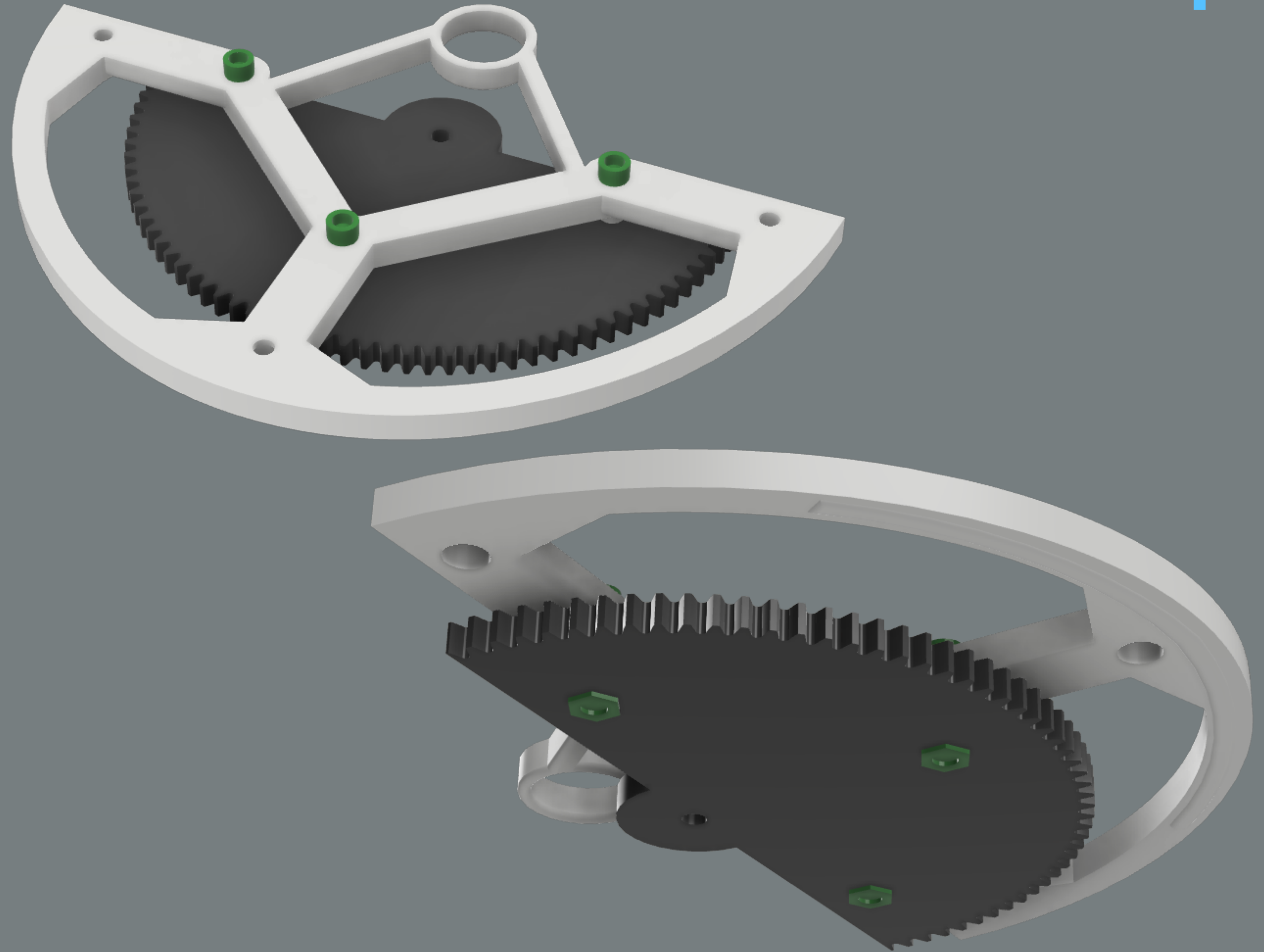


2.5mm Hex Key



****Make sure the screws are fully inserted and don't protrude past the Gearbox Top or else the Neck Mount may hit them.**

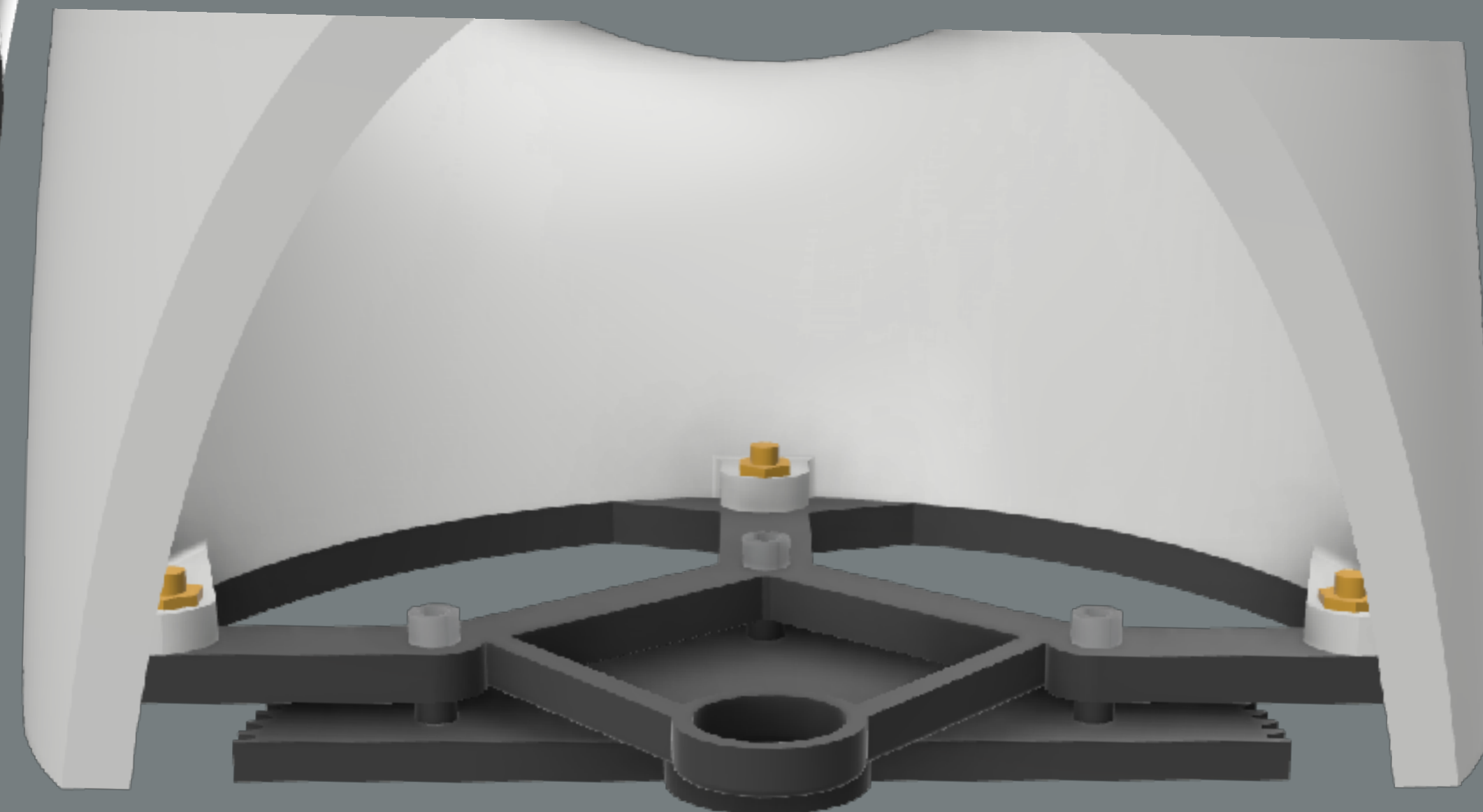
Using **(3)M3x12 Screws** and **(3)M3 Nuts**, attach the Neck Mount to the Big Gear, ensuring these two pieces are tightly joined.



2.5mm Hex Key



Take the Neck Mount/Big Gear Assembly from **Step 11** and using **(3)M3x10 Screws** and **(3)M3 Nuts**, connect it to the Robot Neck.



2.5mm Hex Key



Use the provided
(4)Phillips Screws
to attach the
Raspberry Pi to
the back of the
round LCD
display.

Step 13



Phillips Driver



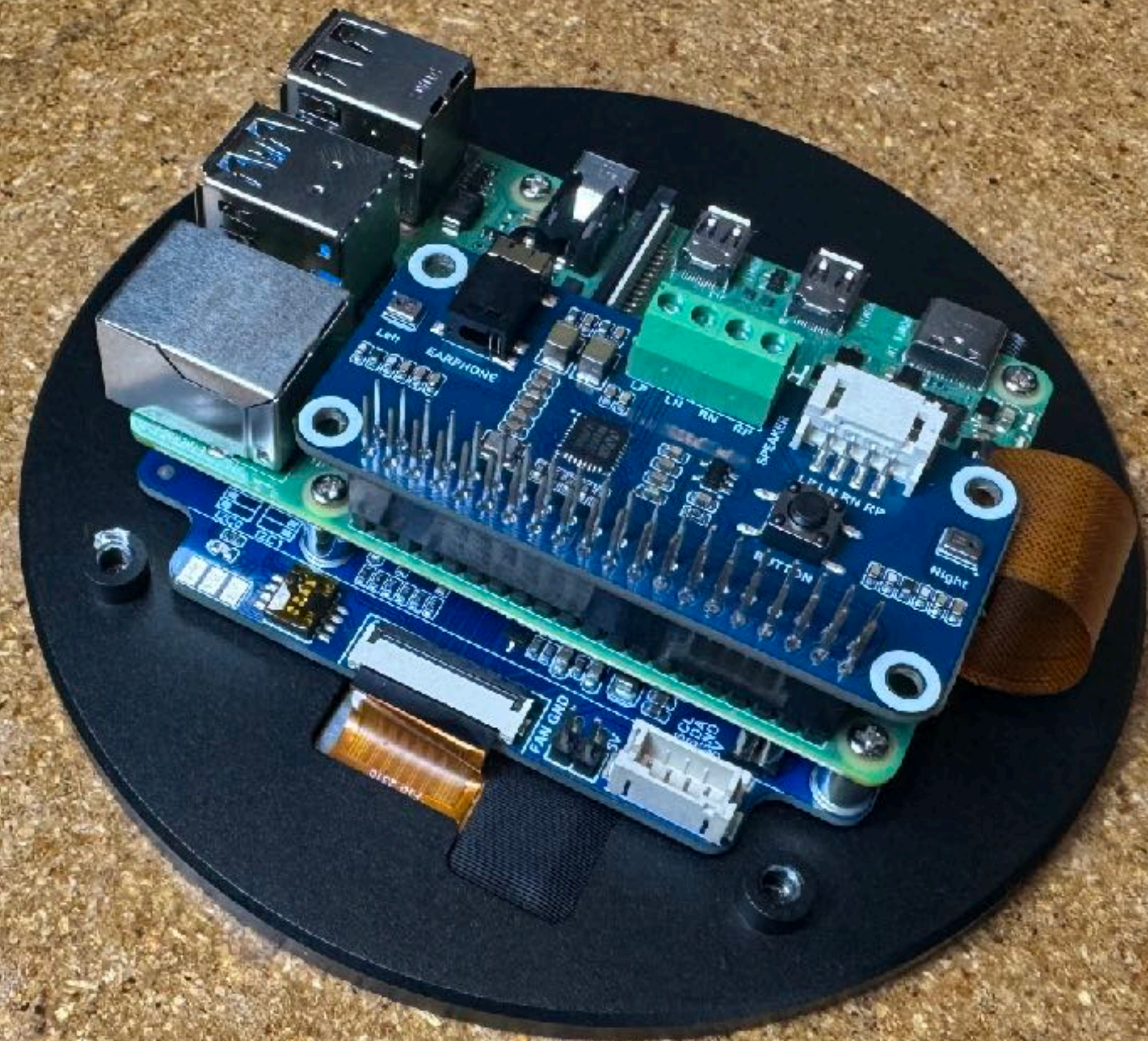
Connect the included **DSI ribbon cable** from the LCD to the corresponding DSI port on the Raspberry Pi.

****** Make sure to insert the cable so the pins are correctly touching



Carefully insert
the **WM8960**
Audio HAT onto
the GPIO header
of the Raspberry
Pi, ensuring all
GPIO pins are
properly aligned.

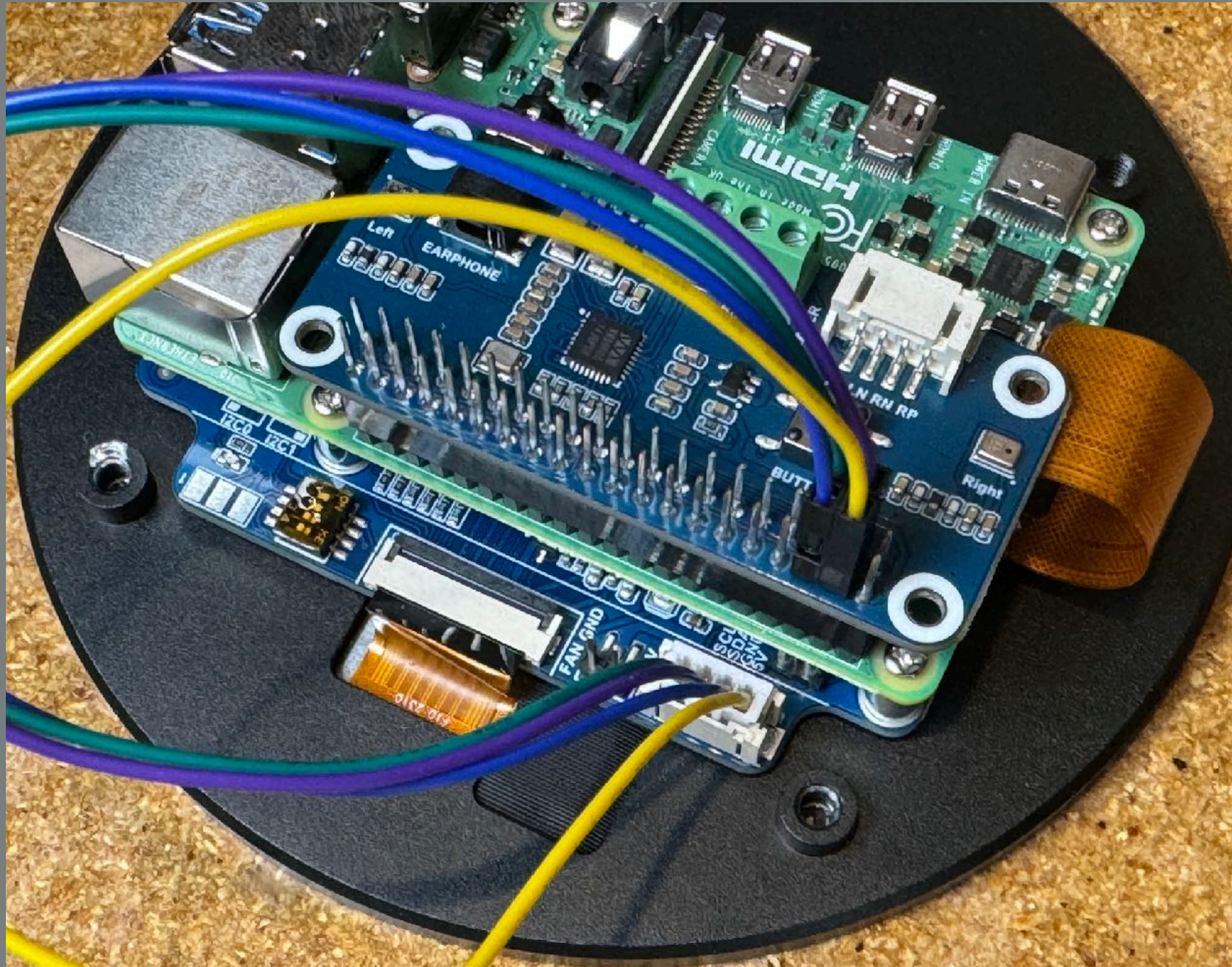
Step 15



Step 15 (Cont'd)

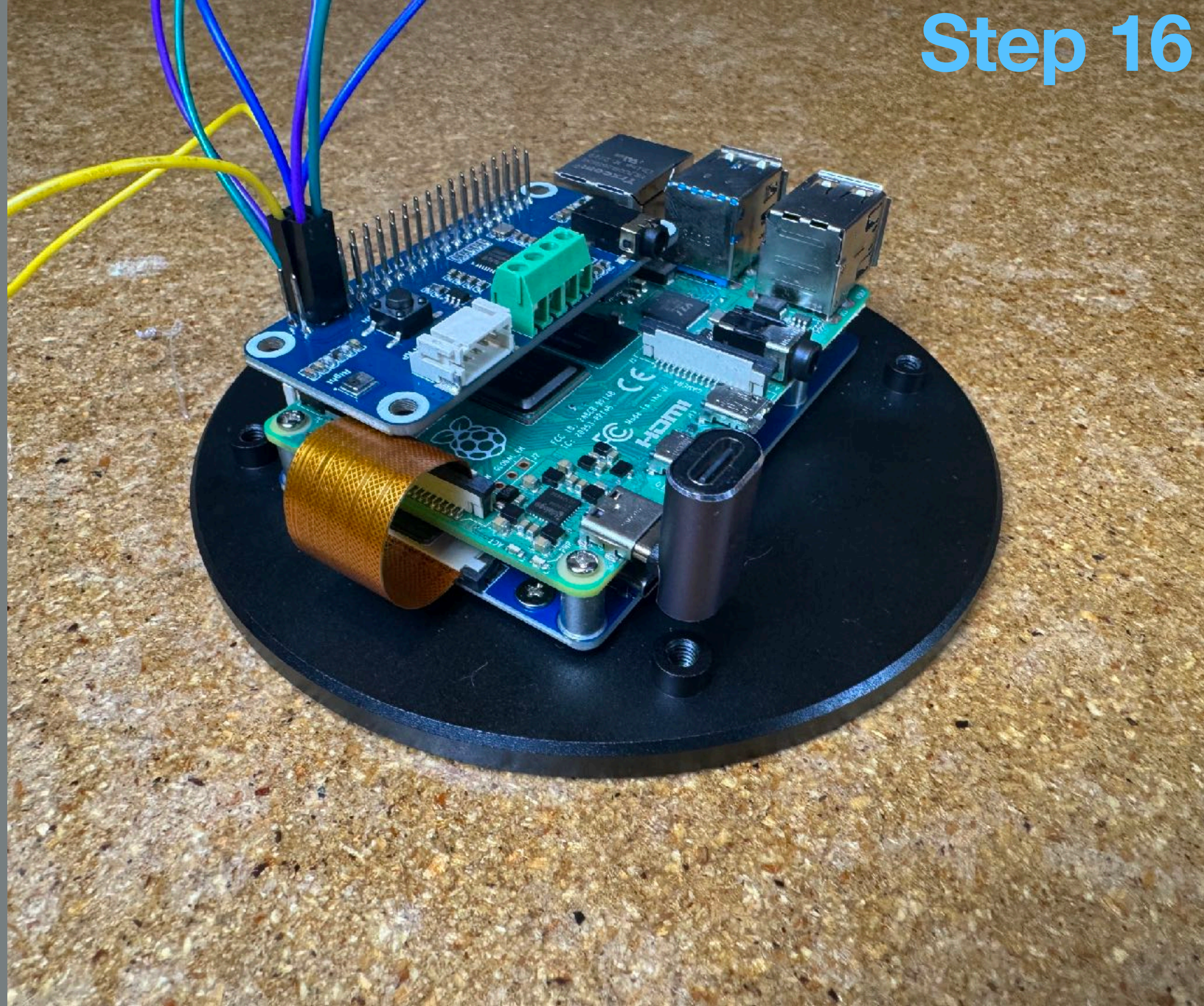
Use the included **4-pin cable** to connect the display to the appropriate GPIO pins on the Raspberry Pi

```
- SDA (Display) --> GPIO 2 (Pin 3) on the Pi  
- 5V (Display)  --> 5V Power (Pin 4) on the Pi  
- SCL (Display) --> GPIO 3 (Pin 5) on the Pi  
- GND (Display) --> Ground (Pin 6) on the Pi
```



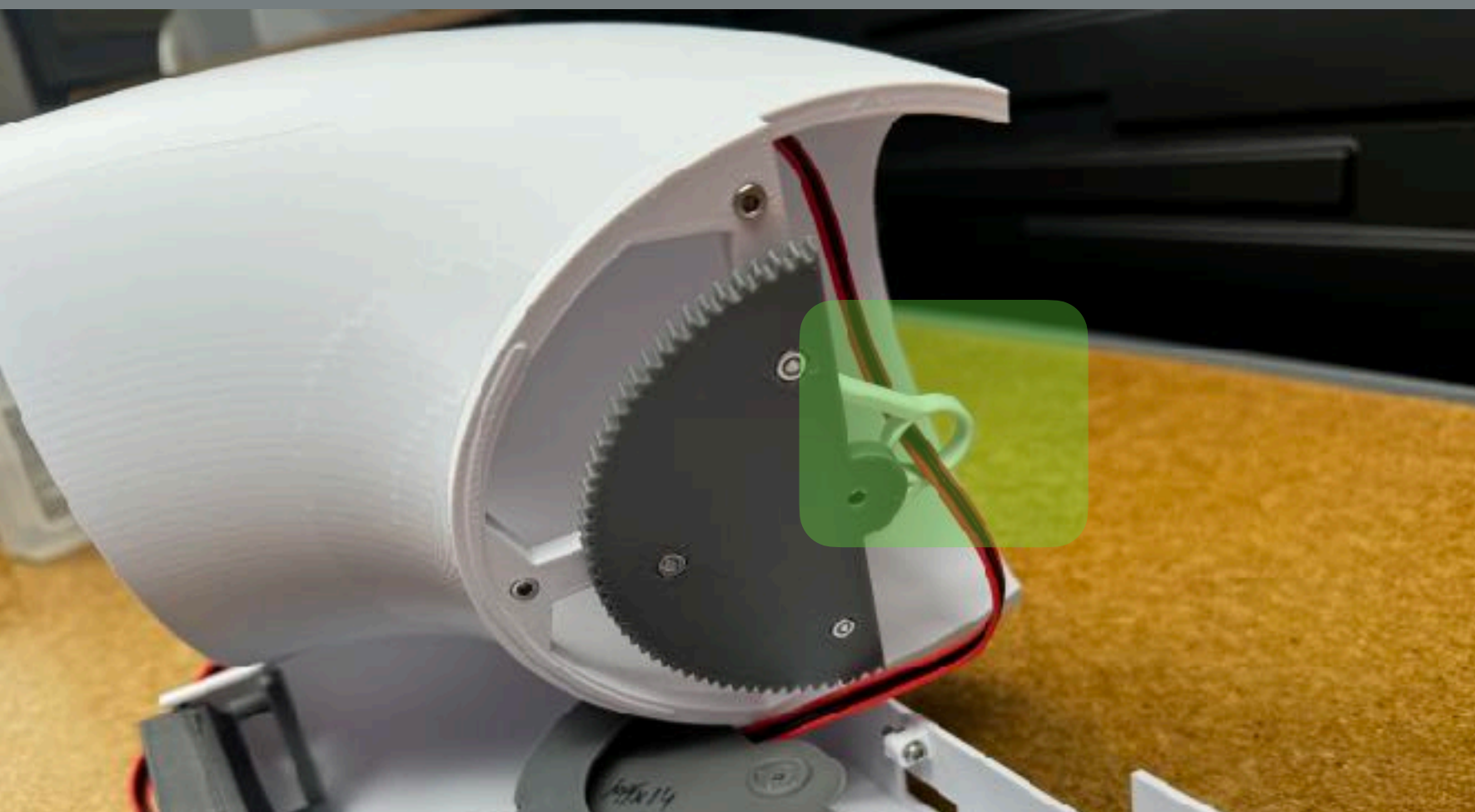
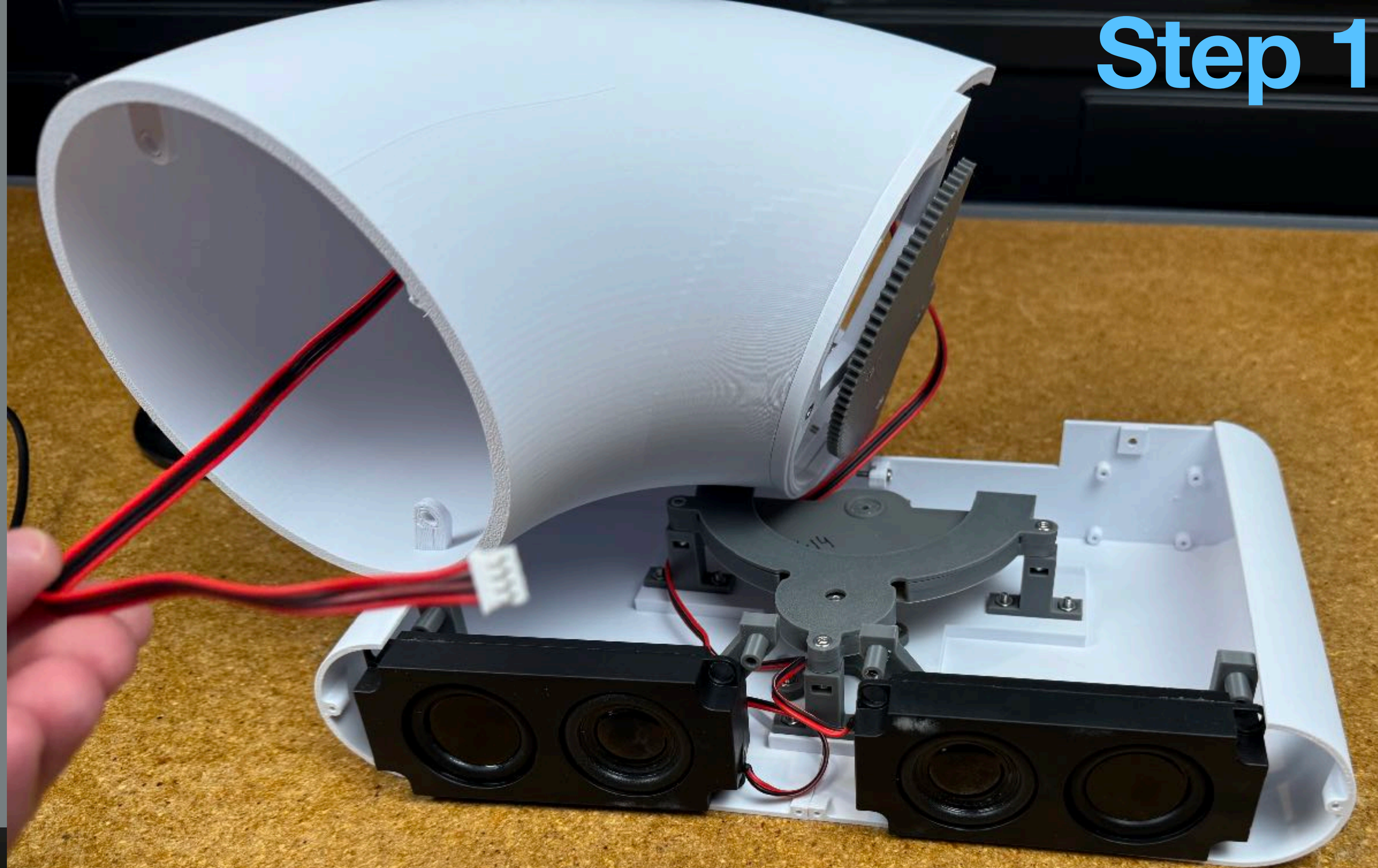
Connect the **right angle USB-C adapter** to the power port on the Pi, this will allow for the Pi's power supply to be plugged in without hitting the neck.

Step 16



Position the two speakers at the front of the case but do not install them. Run the single 4-pin plug from the two speakers through the wiring loop on the Neck Mount, guiding it up through the neck and out the front.

Step 17



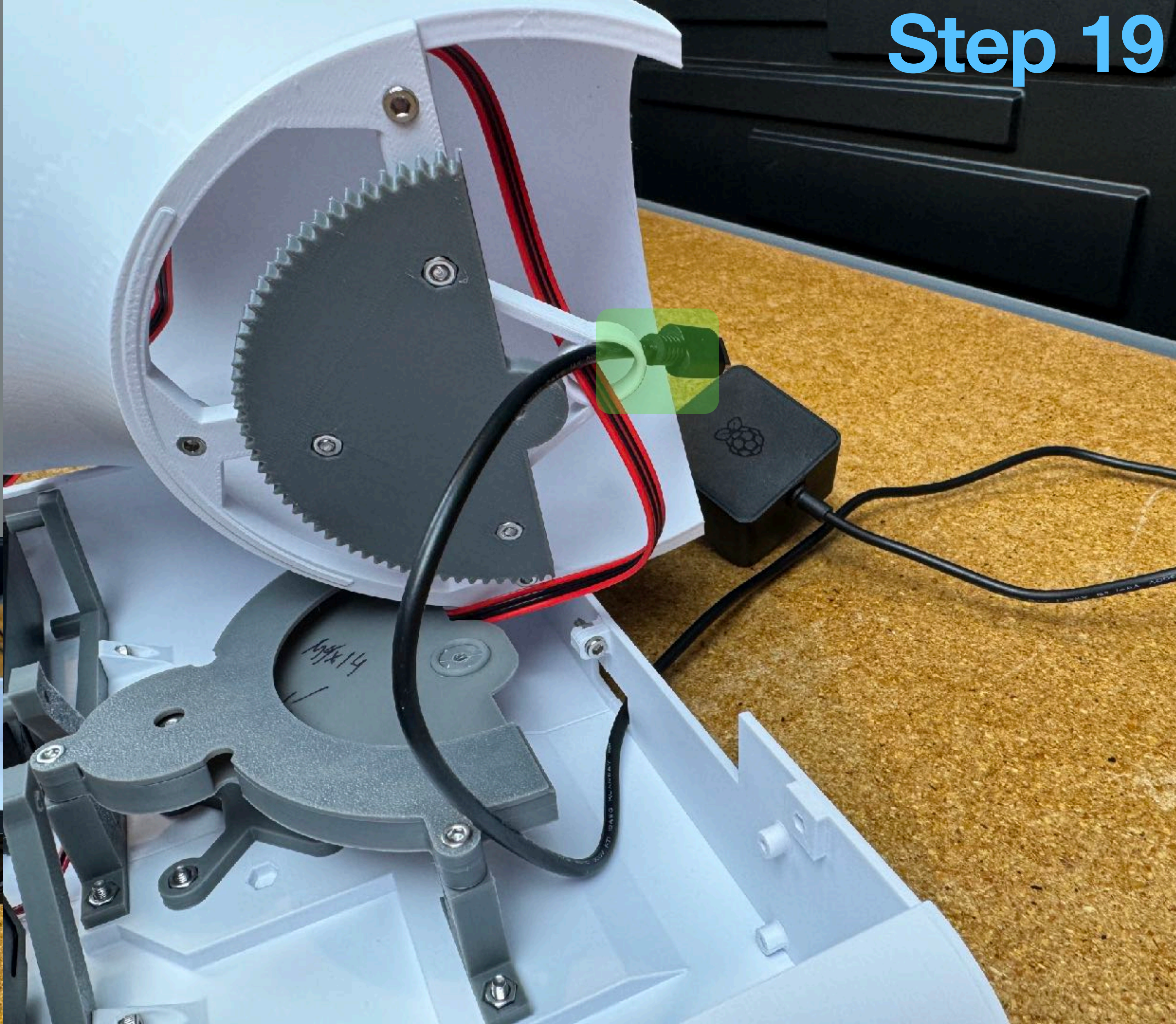
With the plug hanging out from the front of the neck, connect it to the corresponding port on the WM8960 Audio HAT. Be careful not to apply too much pressure when connecting, as this could bend the GPIO pins.

Step 18



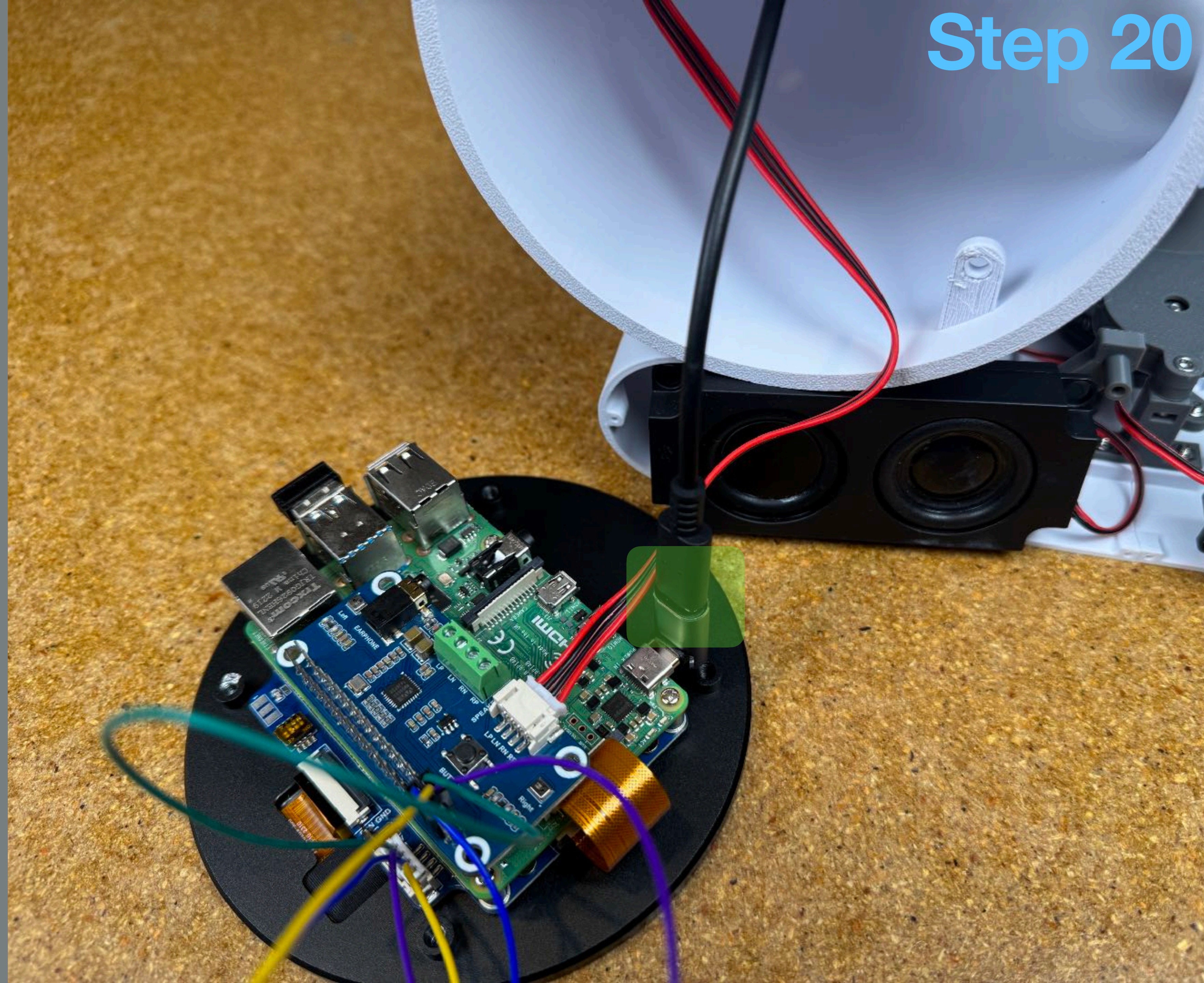
Run the USB-C connection from the Pi's power adapter through the rectangular hole in the back of the case, through the wiring loop on the Neck Mount, guiding it up through the neck and out the front.

Step 19



With the plug hanging out from the front of the neck, connect it to the right angle USB-C adapter that was installed in **Step 16**.

Step 20

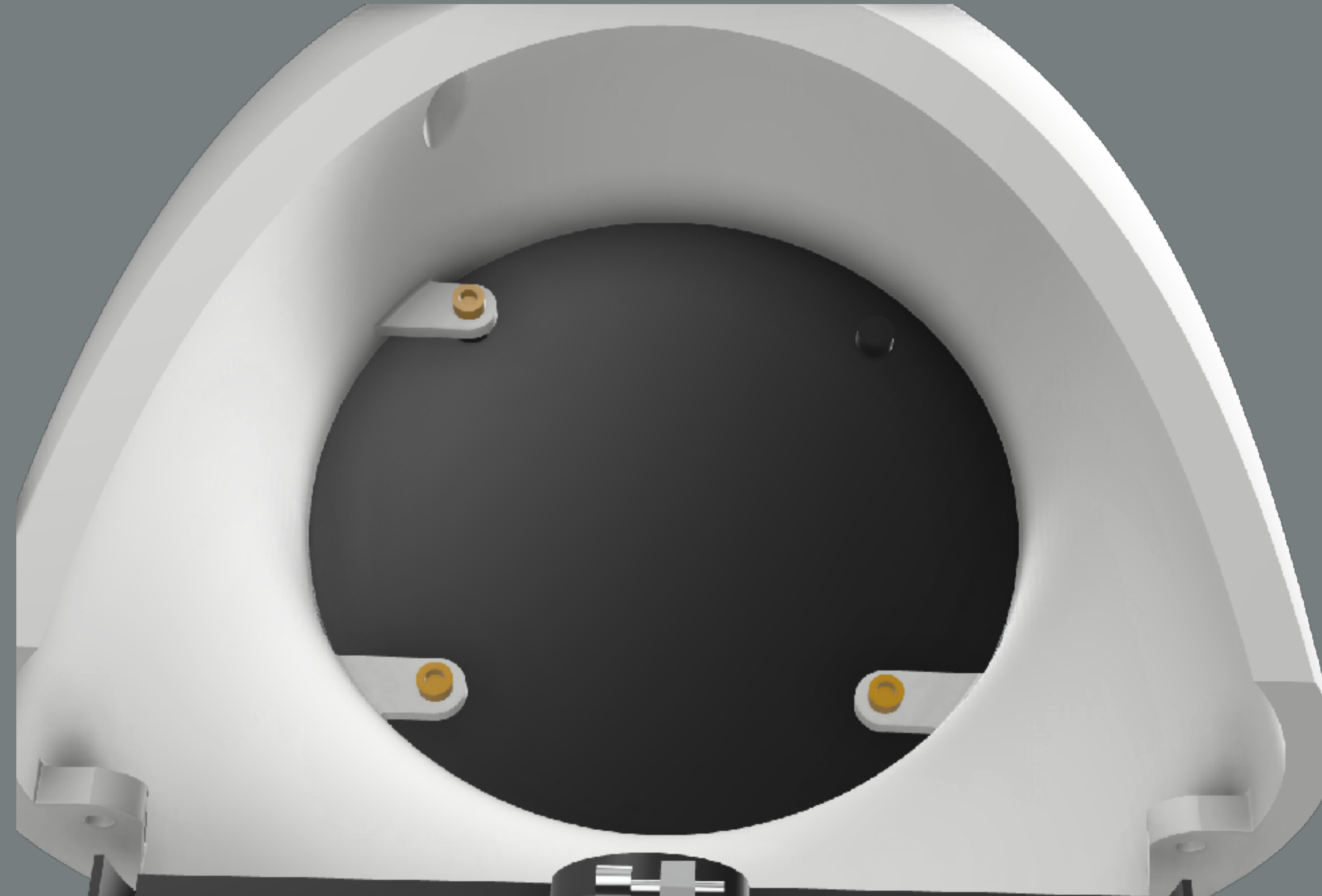


The display assembly has 4 screw holes, but the Neck only has 3. The screw hole located behind the Raspberry Pi USB-C power adapter extension will remain unused, as there is no space to secure a screw there.

To correctly position the display, align it so that the Raspberry Pi's power port is at the corner of the neck without a screw hole.

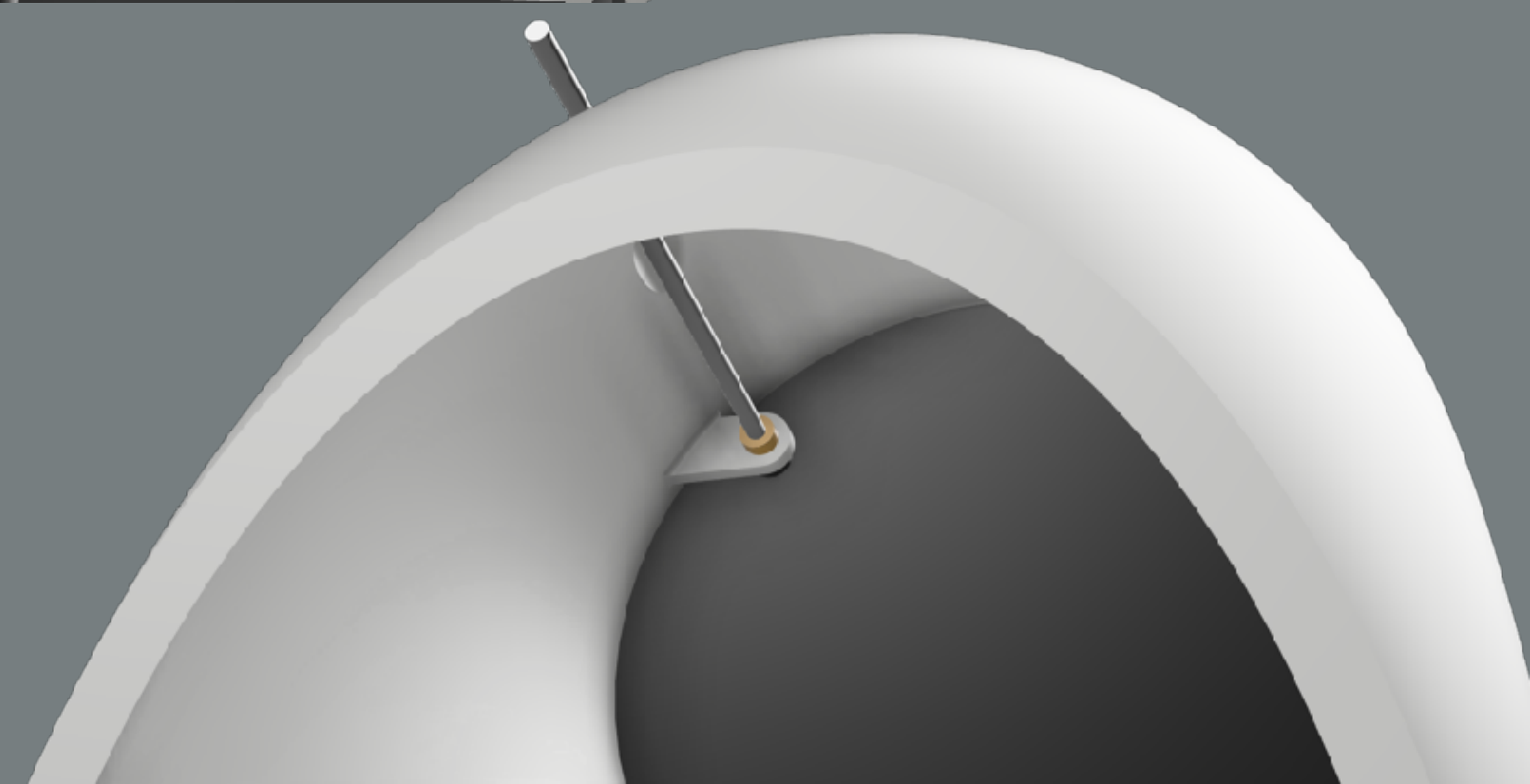


Using **(3) M4x8 Screws**, secure the Display Assembly to the Neck. The bottom two screw holes can be accessed from the back of the neck, while the top screw hole can be reached using a hex key through the access hole at the top back of the neck.



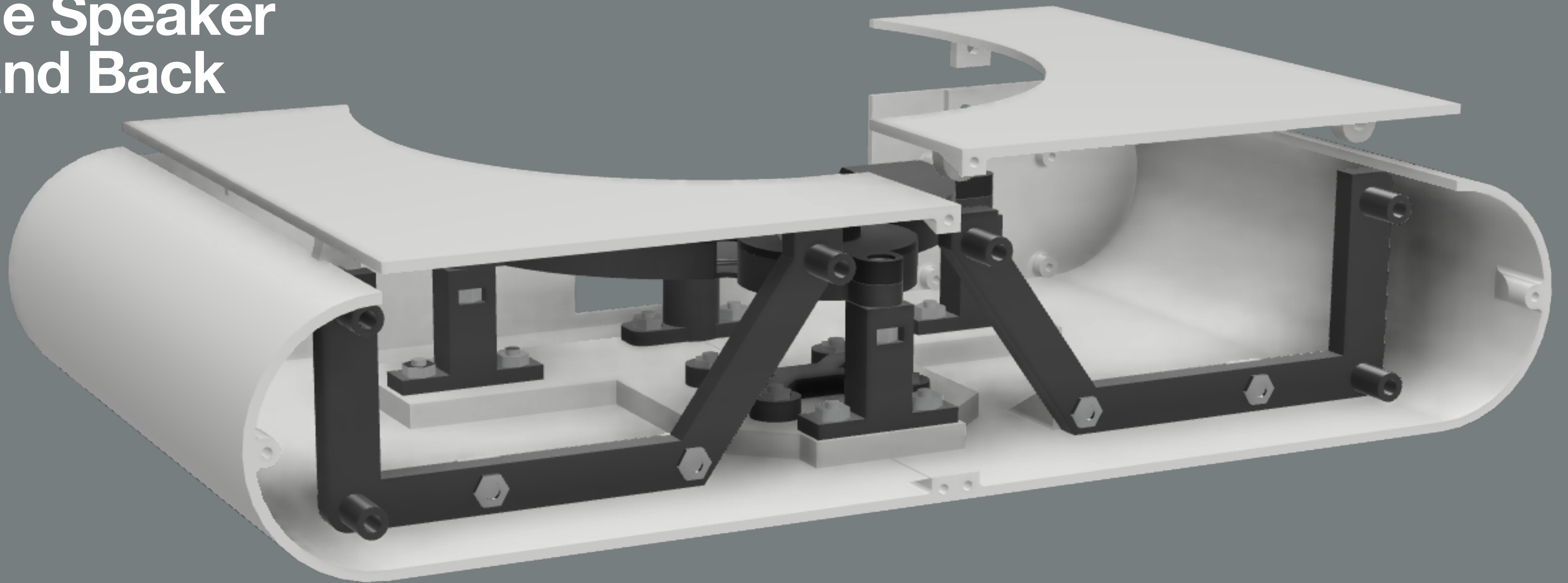
****This step can be a bit tight. Do not fully tighten any screw until all three are in place. If you have large hands, you can detach the Neck Mount from the Neck to get more room inside the neck. If you do this, be sure the wires stay within the wiring loop and re attach it per Step 12.**

3mm Hex Key



**Attach both Case
Top Pieces by
placing them over
the top of the case,
ensuring the tabs
line up correctly
behind the Speaker
Mounts and Back
Case.**

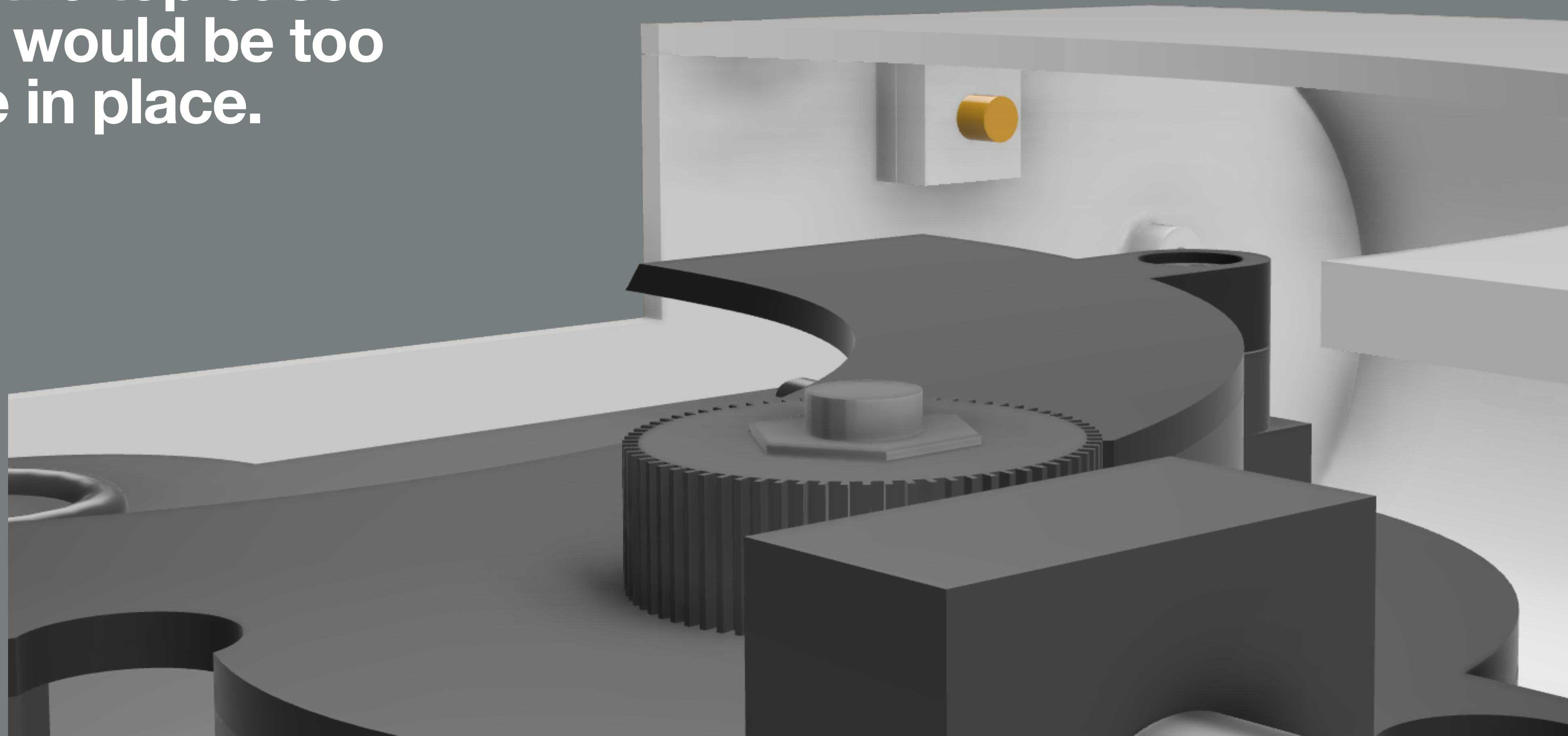
Step 23



Step 24

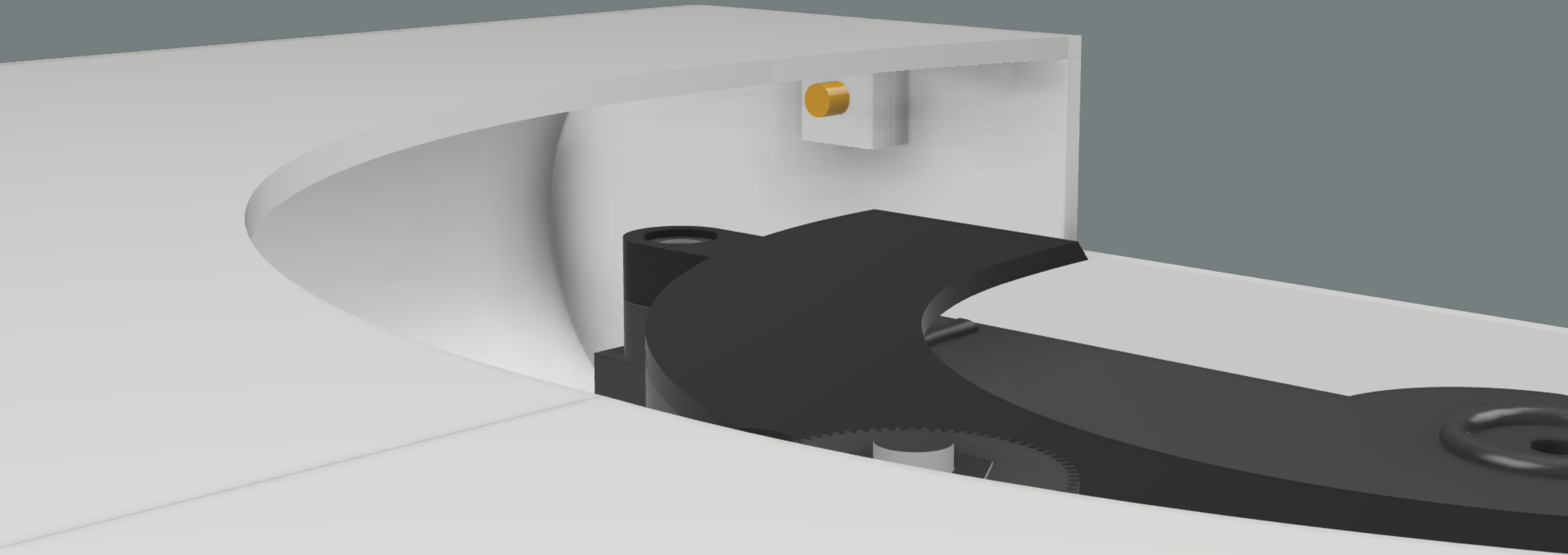
With the Top Case Pieces in place, insert **(1) M3x12 Screw** through the hole in the back of the case piece, through the Top Case Piece, tightening it to secure the back of the Top Case Piece. There are no M3 Nuts holding the top case piece on, as they would be too difficult to secure in place.

2.5mm Hex Key



Repeat The Previous Step for the opposite Case Top Piece.

Step 25



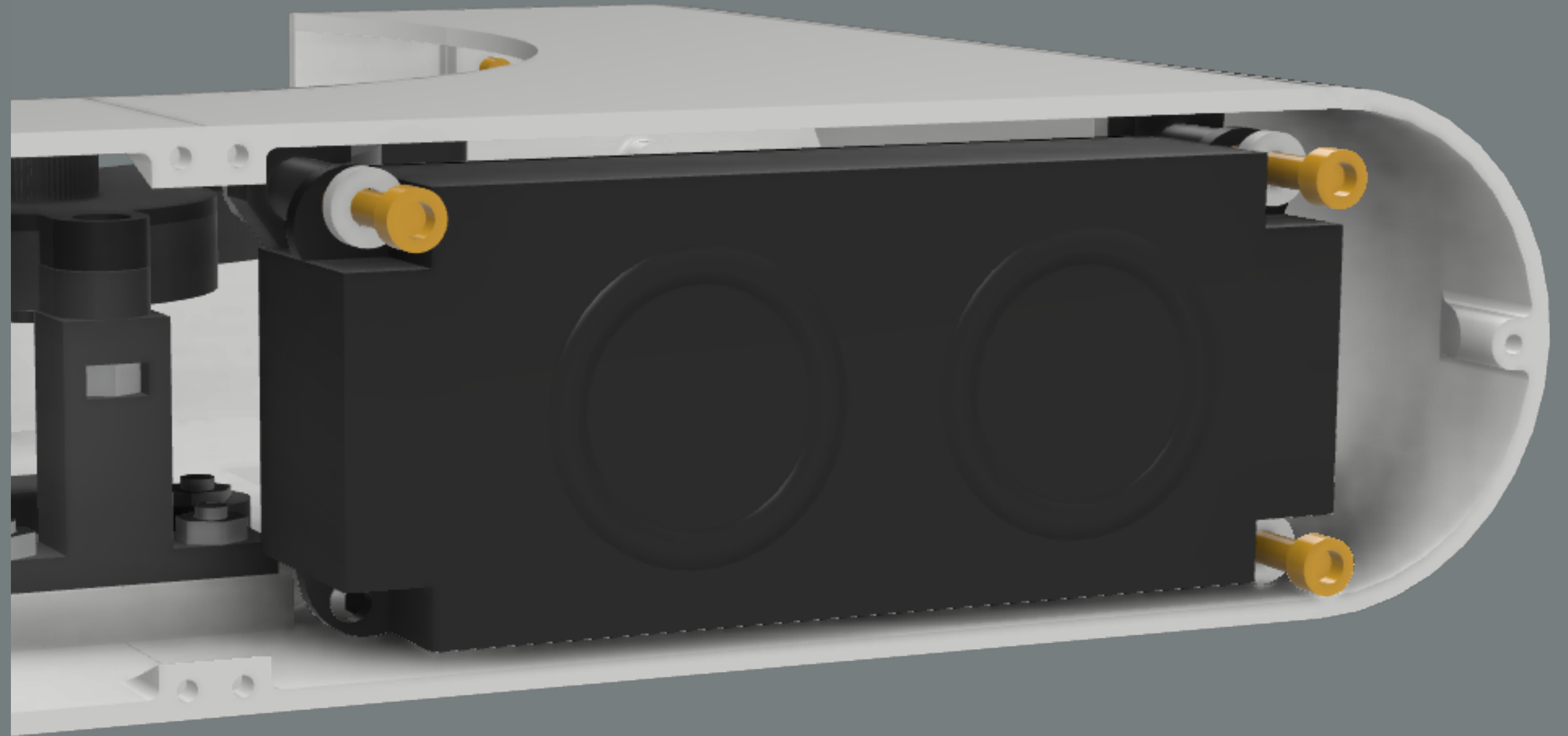
2.5mm Hex Key



Step 26

Attach a speaker to the Speaker Mount using **(3) M3x25 Screws**, **(3) M3 Nuts**, and **3 Plastic washers** placed in front of the speaker tabs.

The top two M3x25 Screws will secure both the speaker, and the Top Case Piece behind it.

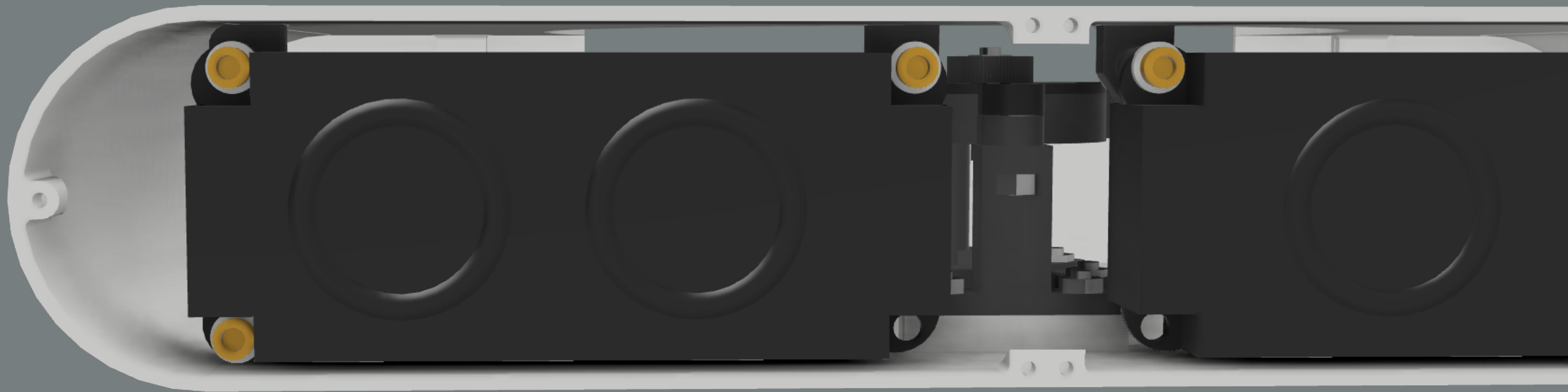


2.5mm Hex Key



Repeat The Previous
Step for the opposite
speaker.

Step 27

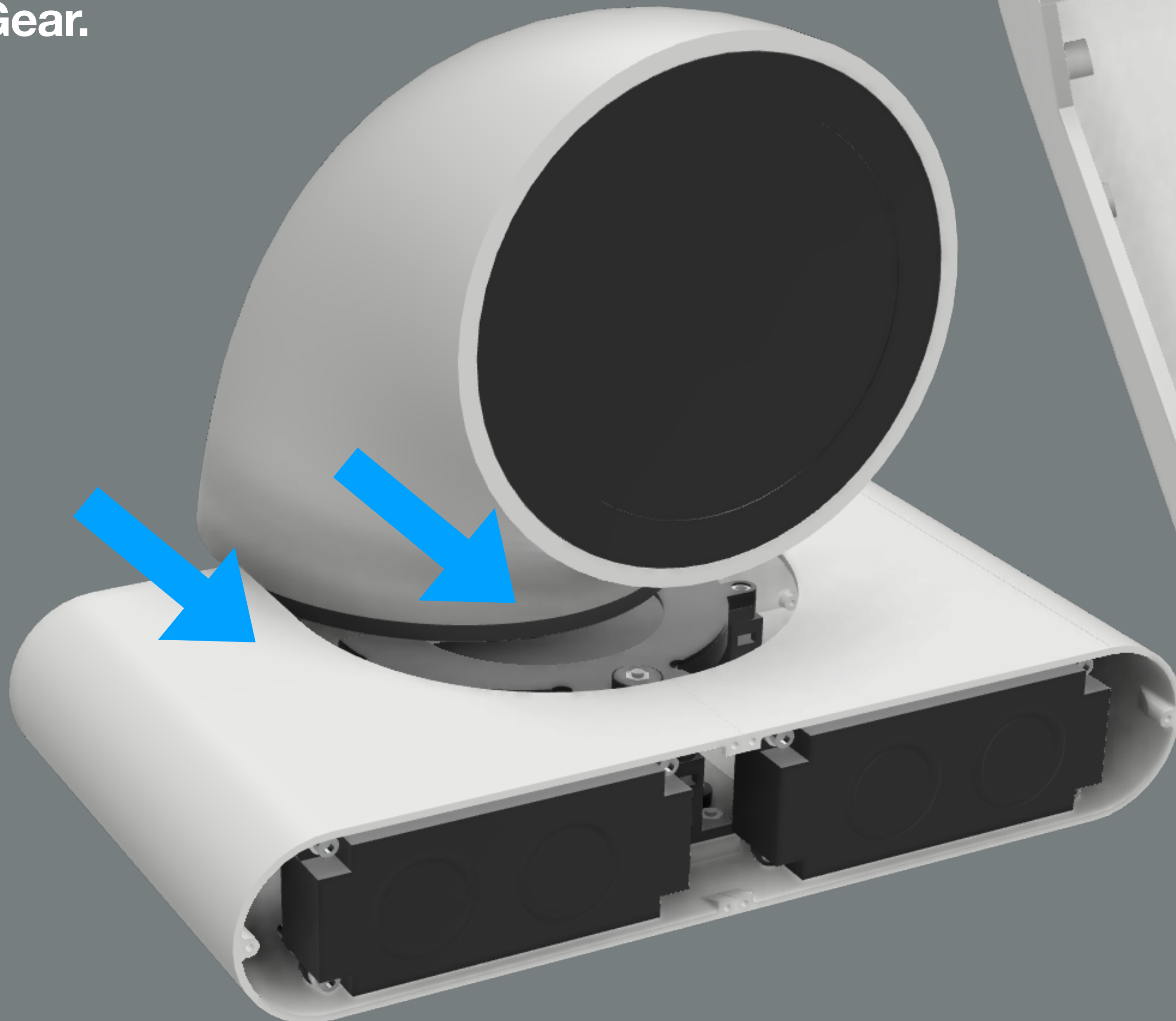


2.5mm Hex Key



Step 28

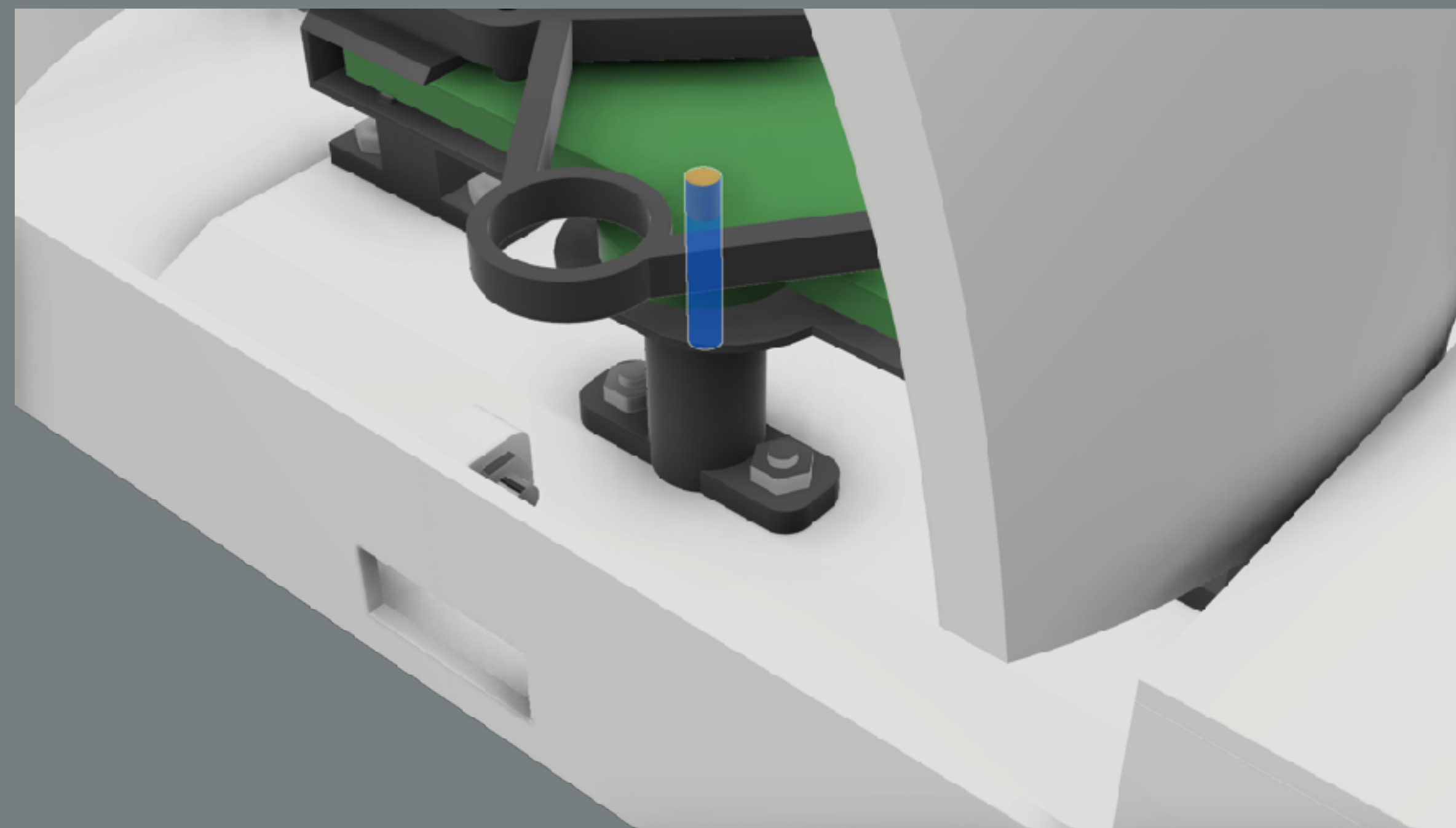
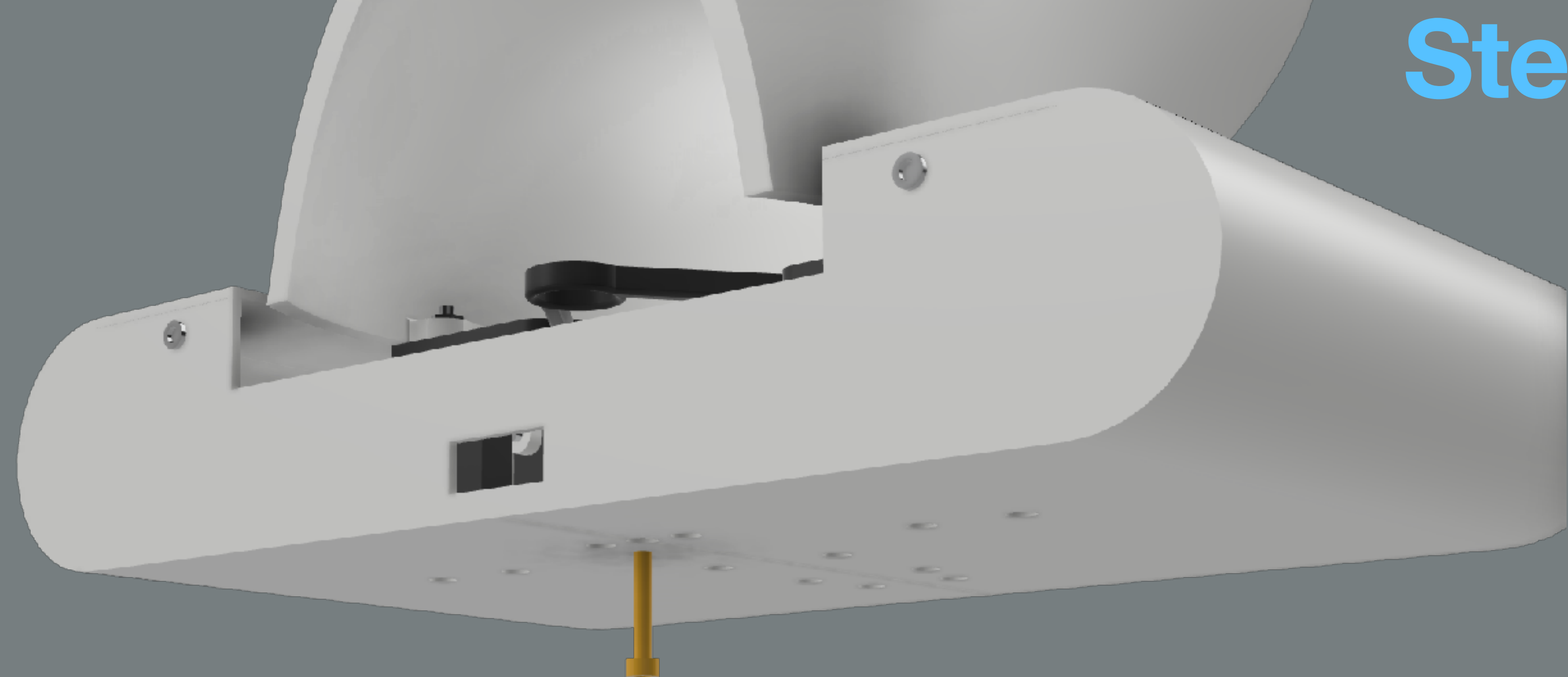
Take the Neck/Display assembly and slide the Big Gear into the Gearbox until it meshes with the Small Gear.



****Note how the Big Gear and Small Gear mesh. The Big Gear goes between the Gearbox Top and Gearbox Bottom, while the Neck Mount, rests on top of the Gearbox Top.**

Step 29

To fully mate the Neck/Display with the rotation mechanism, insert **(1) M3x20 Screw** through the bottom of the case, through the Round Gearbox Post, through the hole in the Gearbox Bottom, and through the hole in the Big Gear. This will keep the Neck/Display assembly aligned with the rotation mechanism and case.



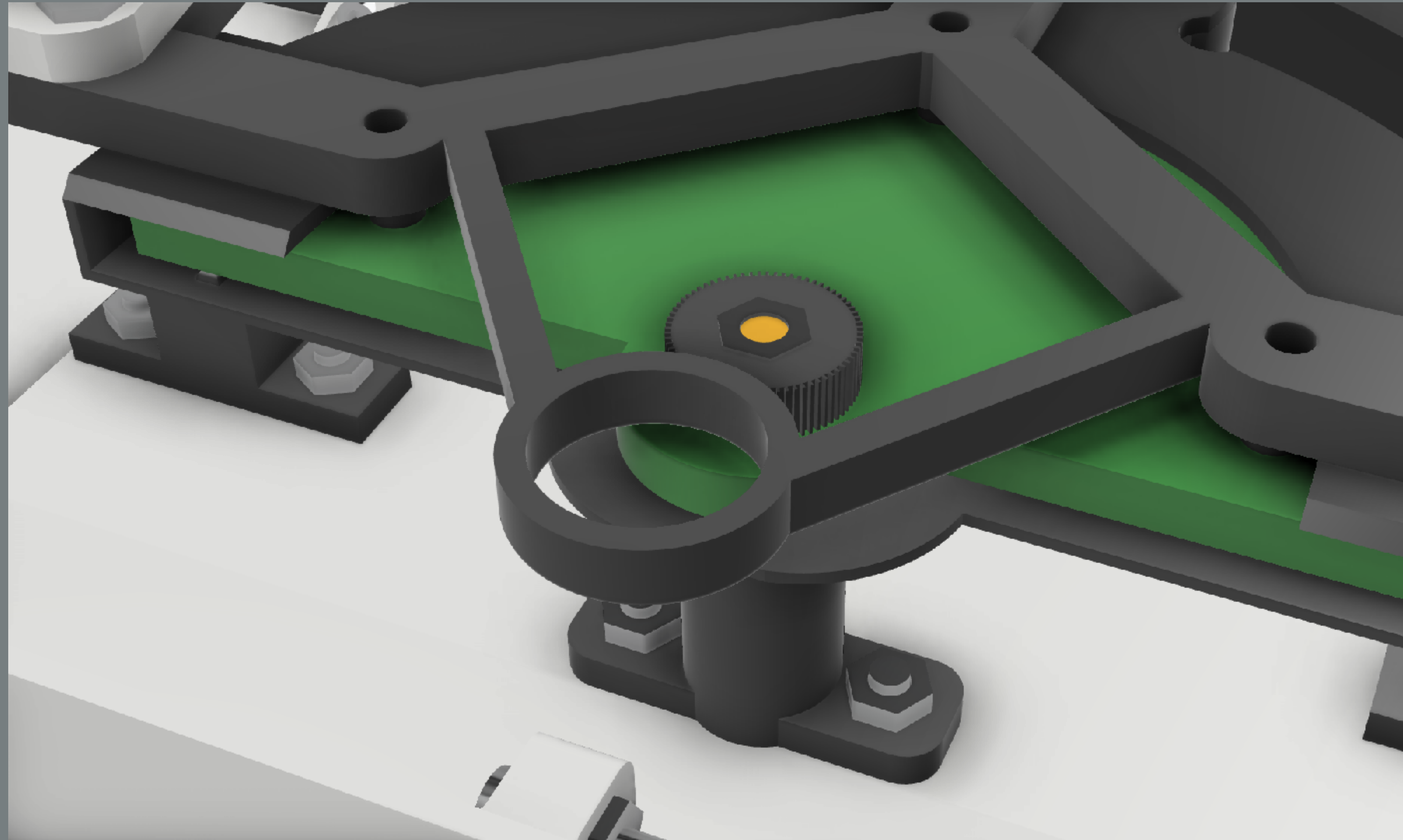
2.5mm Hex Key



Place **(1) M3 Nut** inside the Gear Top Cover, and tighten it over the **M3x20 Screw** sticking out of the Big Gear. Do not over tighten this Nut or you will not be able to spin the rotation mechanism.

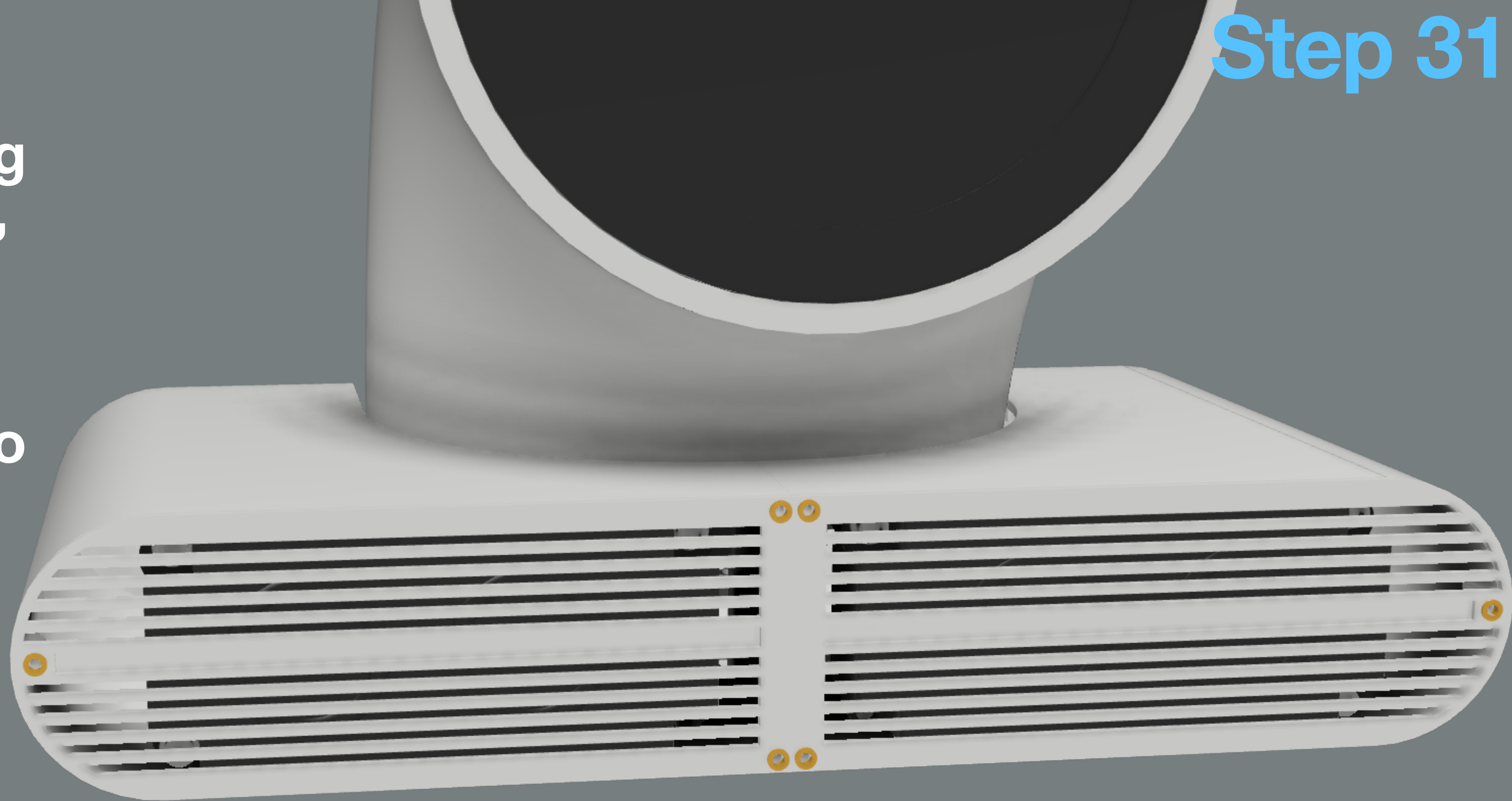
This will keep the Neck/Display assembly attached to the rotation mechanism and case

2.5mm Hex Key



Step 31

Take the two Speaker Grill Pieces and using **(6) M2x4 Screws**, screw them into the holes in the case assembly opposite them to complete the assembly.



2mm Hex Key



This Concludes the build instructions